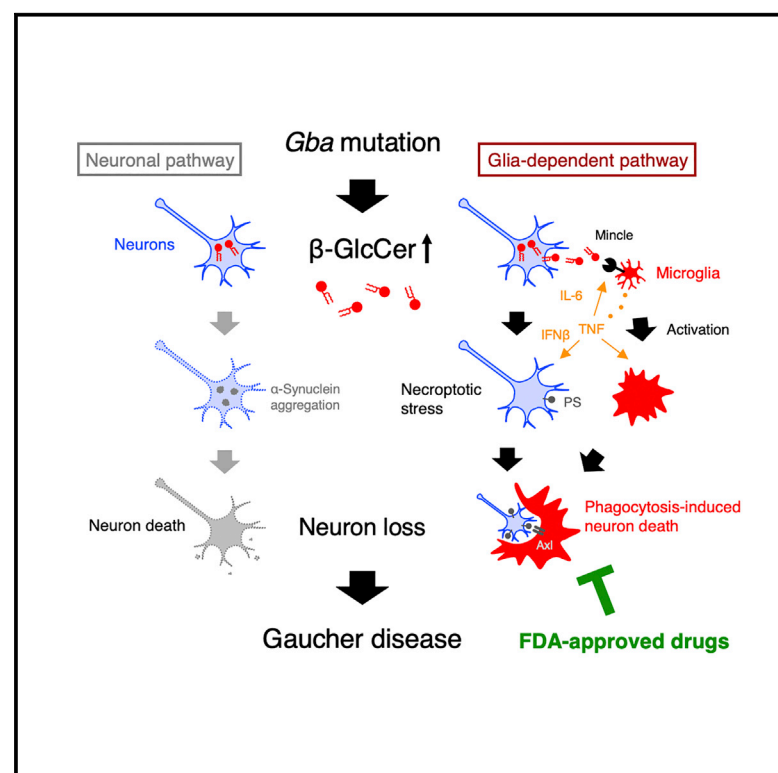


Direct activation of microglia by β -glucosylceramide causes phagocytosis of neurons that exacerbates Gaucher disease

Graphical abstract



Authors

Takashi Shimizu, Charles R. Schutt, Yoshihiro Izumi, ..., Atsushi Kumanogoh, Masaki Takao, Sho Yamasaki

Correspondence

yamasaki@biken.osaka-u.ac.jp

In brief

Gaucher disease is an inherited, incurable disorder with neuropathological complications caused by β -GlcCer accumulation, but its molecular action is unclear. Shimizu et al. show that β -GlcCer directly activates microglia to promote neuroinflammation leading to phagocytosis of living neurons and blockade of this pathway with FDA-approved drugs improves symptoms and survival.

Highlights

- β -GlcCer directly activates microglia through Mincle in GD
- Neuronal loss caused by microglial phagocytosis exacerbates GD
- Microglia-induced necroptotic stress sensitizes neurons for phagocytosis
- Blocking the vicious cycle, shared in human GD, by repurposed drugs remedied symptoms



Article

Direct activation of microglia by β -glucosylceramide causes phagocytosis of neurons that exacerbates Gaucher disease

Takashi Shimizu,^{1,2} Charles R. Schutt,¹ Yoshihiro Izumi,³ Noriyuki Tomiyasu,³ Zakaria Omahdi,^{1,2} Kuniyuki Kano,⁴ Hyota Takamatsu,^{5,6} Junken Aoki,⁴ Takeshi Bamba,³ Atsushi Kumanogoh,^{5,6,7,8} Masaki Takao,⁹ and Sho Yamasaki^{1,2,10,*}

¹Department of Molecular Immunology, Research Institute for Microbial Diseases, Osaka University, Suita, Osaka 565-0871, Japan

²Laboratory of Molecular Immunology, Immunology Frontier Research Center (IFReC), Osaka University, Suita, Osaka 565-0871, Japan

³Division of Metabolomics, Research Center for Transomics Medicine, Medical Institute of Bioregulation, Kyushu University, Fukuoka, Fukuoka 812-8582, Japan

⁴Department of Health Chemistry, Graduate School of Pharmaceutical Sciences, The University of Tokyo, Bunkyo-ku, Tokyo 113-0033, Japan

⁵Department of Respiratory Medicine and Clinical Immunology, Osaka University Graduate School of Medicine, Suita, Osaka 565-0871, Japan

⁶Department of Immunopathology, Immunology Frontier Research Center (IFReC), Osaka University, Suita, Osaka 565-0871, Japan

⁷Integrated Frontier Research for Medical Science Division, Institute for Open and Transdisciplinary Research Initiatives (OTRI), Osaka University, Suita, Osaka 565-0871, Japan

⁸Center for Infectious Diseases for Education and Research (CiDER), Osaka University, Suita, Osaka 565-0871, Japan

⁹Department of Clinical Laboratory, National Center of Neurology and Psychiatry, Kodaira, Tokyo 187-8551, Japan

¹⁰Lead contact

*Correspondence: yamasaki@biken.osaka-u.ac.jp

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SUMMARY

Gaucher disease (GD) is the most common lysosomal storage disease caused by recessive mutations in the degrading enzyme of β -glucosylceramide (β -GlcCer). However, it remains unclear how β -GlcCer causes severe neuronopathic symptoms, which are not fully treated by current therapies. We herein found that β -GlcCer accumulating in GD activated microglia through macrophage-inducible C-type lectin (Mincle) to induce phagocytosis of living neurons, which exacerbated Gaucher symptoms. This process was augmented by tumor necrosis factor (TNF) secreted from activated microglia that sensitized neurons for phagocytosis. This characteristic pathology was also observed in human neuronopathic GD. Blockade of these pathways in mice with a combination of FDA-approved drugs, minocycline (microglia activation inhibitor) and etanercept (TNF blocker), effectively protected neurons and ameliorated neuronopathic symptoms. In this study, we propose that limiting unrestrained microglia activation using drug repurposing provides a quickly applicable therapeutic option for fatal neuronopathic GD.

INTRODUCTION

Germline-encoded pattern recognition receptors (PRRs) have been identified as receptor families that sense characteristic molecular features of pathogens.¹ Recently, PRRs have been also found to recognize various endogenous molecules² and thus are considered a broad sensor for “perturbations” from external or internal stress.² Since the balanced regulation of dynamic cellular metabolism is critical for tissue homeostasis, one of the typical internal stresses is the aberrant accumulation of cellular components, such as proteins, lipids, carbohydrates, or other metabolites.³ Indeed, inherited metabolic dysfunction leads to accumulation of metabolites in the lysosomes, an organelle responsible for proper degradation and causes disorders called lysosomal storage diseases (LSDs).⁴ However, it is

unclear whether the immune system influences tissue homeostasis through PRRs recognizing these accumulated metabolites in physiological or pathological settings.

Gaucher disease (GD) is an LSD caused by autosomal recessive mutations in glucocerebrosidase (GCase, *GBA*), an enzyme that degrades β -glucosylceramide (β -GlcCer). As a consequence of β -GlcCer accumulation, GD patients exhibit typical symptoms, such as hematological, visceral, skeletal, and neurological abnormalities, but the pathogenesis of β -GlcCer is not fully understood.⁵ Current first-line treatment is enzyme replacement therapy (ERT)⁶; however, the effectiveness of ERT in treating the neuronopathic symptoms in GD is limited,⁶ and different targets in the brain need to be identified. Although β -GlcCer or a related metabolite cause intrinsic toxicity in neurons,⁷ the extrinsic effects by surrounding cells remain unclear. Mouse



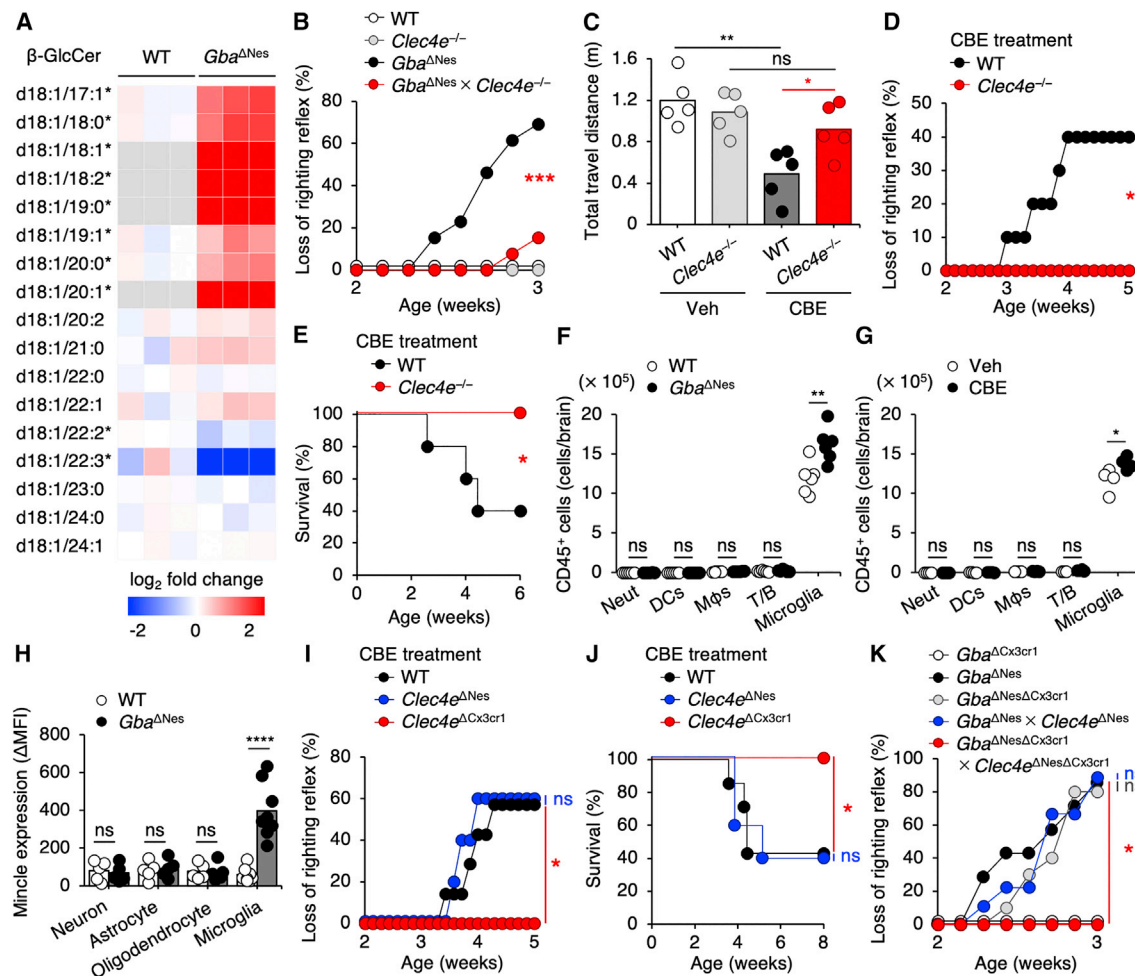


Figure 1. Contribution of Mincle in genetic and pharmacological GD models

(A) Heatmap showing log₂-fold change in the amount of β -GlcCer species in the brain of P10 $Gba^{\Delta Nes}$ mice compared with P10 WT mice (n = 3 mice per group). (B) Percentage of loss of righting reflex of WT (n = 10), $Clec4e^{-/-}$ (n = 10), $Gba^{\Delta Nes}$ (n = 27), and $Gba^{\Delta Nes} \times Clec4e^{-/-}$ (n = 13) mice. (C) Total traveled distance of WT and $Clec4e^{-/-}$ mice treated with or without CBE in the open field test (n = 5 mice per group). (D) Percentage of loss of righting reflex of WT and $Clec4e^{-/-}$ mice treated with CBE (n = 10 mice per group). (E) Kaplan-Meier survival curve of WT and $Clec4e^{-/-}$ mice treated with CBE (n = 5 mice per group). (F) Number of neutrophils (CD45^{hi}TMEM119⁻CD11b⁺Ly-6G⁺), dendritic cells (CD45^{hi}TMEM119⁻CD11b⁺Ly-6G⁻), macrophages (CD45^{hi}TMEM119⁻CD11b⁺Ly-6G⁺F4/80⁺), lymphocytes (CD45^{hi}TMEM119⁻CD11b⁺Ly-6G⁻), and microglia (CD45^{int}TMEM119⁺) in the brain of P21 WT and $Gba^{\Delta Nes}$ mice (n = 6 mice per group). (G) Number of neutrophils, dendritic cells, macrophages, lymphocytes, and microglia as defined in (F) in the brain of P21 WT mice with or without CBE treatment (n = 5 mice per group). (H) Mincle expression among neurons and glial cells in P21 WT and $Gba^{\Delta Nes}$ mice. Data are shown as delta mean fluorescence intensity (Δ MFI), calculated by subtracting MFI of isotype control from MFI of Mincle (neuron, astrocyte, and oligodendrocyte, n = 5; microglia, n = 8). (I) Percentage of loss of righting reflex of WT (n = 7), $Clec4e^{\Delta Nes}$ (n = 5), and $Clec4e^{\Delta Cx3cr1}$ (n = 6) mice treated with CBE. (J) Kaplan-Meier survival curve of WT (n = 7), $Clec4e^{\Delta Nes}$ (n = 5), and $Clec4e^{\Delta Cx3cr1}$ (n = 6) mice treated with CBE. (K) Percentage of loss of righting reflex of $Gba^{\Delta Cx3cr1}$ (Gba -floxed $\times$ $Cx3cr1$ -cre, n = 11), $Gba^{\Delta Nes}$ (Gba -floxed $\times$ Nes -cre, n = 7), $Gba^{\Delta Nes} \times Cx3cr1$ (Gba -floxed $\times$ Nes -cre and $Cx3cr1$ -cre, n = 10), $Gba^{\Delta Nes} \times Clec4e^{\Delta Nes}$ (Gba -floxed and $Clec4e$ -floxed $\times$ Nes -cre, n = 9), and $Gba^{\Delta Nes} \times Cx3cr1 \times Clec4e^{\Delta Nes}$ (Gba -floxed and $Clec4e$ -floxed $\times$ Nes -cre and $Cx3cr1$ -cre, n = 4) mice.

Data are representative of two independent experiments (D and E), each dot represents an individual mouse (C and F–H), and bars represent the mean of each group (C and H). The log-rank test (B, D, E, and I–K), the one-way ANOVA followed by Tukey–Kramer test (C), and the unpaired two-tailed Student's t test (A and F–H) were used for statistical analysis. *p < 0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001; ns, not significant.

See also Figure S1 and Table S1.

models, such as genetic models bearing mutant Gba alleles or drug-inducible models using GCase inhibitors⁸ are useful to decipher the pathogenic mechanisms as they recapitulate β -GlcCer accumulation, neuronal loss, motor dysfunction, and delayed growth.⁸

We have recently reported that macrophage-inducible C-type lectin (Mincle, $Clec4e$) is a receptor for β -GlcCer derived from damaged cells as a danger-associated molecular pattern (DAMP).⁹ As Mincle is also expressed on microglia,¹⁰ this β -GlcCer receptor may contribute to the homeostatic immune

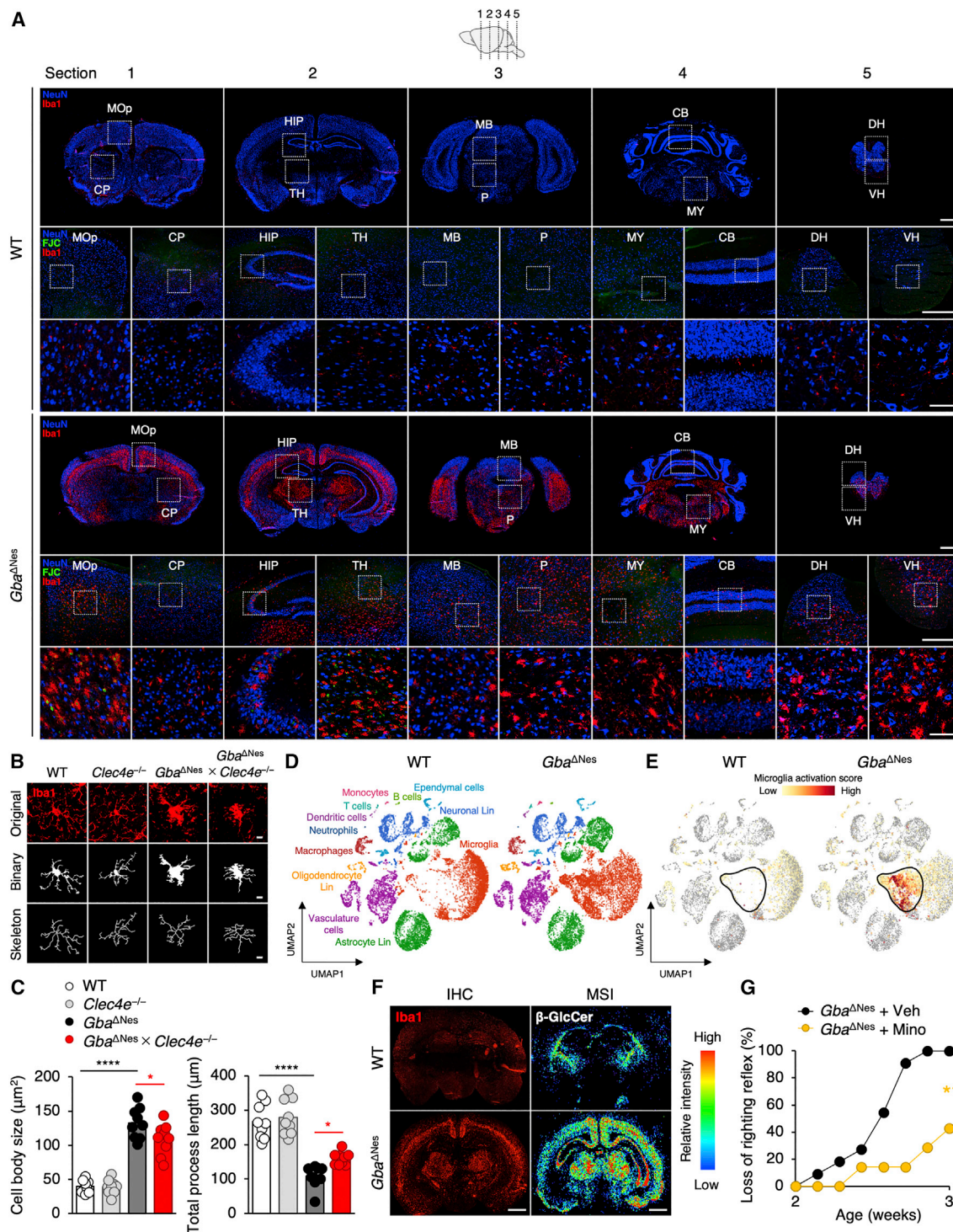


Figure 2. Role of microglia activation in *Gba*^{ΔNes} mice

(A) Coronal sections of P21 brain from WT and *Gba*^{ΔNes} mice corresponding to the sections depicted in the schematic stained with NeuN (blue), FJC (green), and Iba1 (red). MOp, primary motor cortex; CP, caudate putamen; HIP, hippocampus; TH, thalamus; MB, midbrain; P, pons; MY, medulla; CB, cerebellum; DH, dorsal horn; VH, ventral horn. For each genotype, three sections/mouse from three mice were analyzed and representative images were shown.

(B) Representative Iba1-stained microglia (red) with binary and skeletonized images from cerebral cortex of P21 WT, *Clec4e*^{-/-}, *Gba*^{ΔNes}, and *Gba*^{ΔNes} × *Clec4e*^{-/-} mice from three independent experiments.

(C) Iba1⁺ cell body area (left) and total process length (right). Three cells per mouse from three mice. Bars represent the mean of each group.

(D) UMAP plots show clustering of brain cells from P14 WT and *Gba*^{ΔNes} mice.

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responses by sensing physiological stresses in the brain. However, excessive accumulation of β -GlcCer, as in GD, may induce dysregulated activation of microglia, which potentially leads to brain dysfunction.

Here, we investigated the mechanism of β -GlcCer pathology in both mice and humans. Our results revealed that unrestrained activation of microglia by β -GlcCer promotes phagocytosis of living neurons, thereby contributing to neuronal loss in GD. We also identified an effective combination of Food and Drug Administration (FDA)-approved drugs that target this pathway in GD models.

RESULTS

Mincle deficiency ameliorates GD symptoms

To establish a mouse model with features close to neurodegenerative GD, we first deleted GCase in neurons by targeting exons 8 and 9 (*Gba^{flox/flox} × Nes-cre*; *Gba^{ΔNes}*) (Figures S1A–S1C) where mutations are frequently detected in patients¹¹ using *Nes-cre* mice which is expressed in the progenitor cells of neurons, astrocytes, and oligodendrocytes, but not microglia.¹² Quantitative analysis of lipids in the brain of *Gba^{ΔNes}* mice using supercritical fluid chromatography coupled with triple quadrupole mass spectrometry (SFC/MS/MS) revealed the accumulation of major species of β -GlcCer and β -GlcSph (Figure 1A; Table S1). *Gba^{ΔNes}* mice exhibited severe neurological phenotypes, such as loss of righting reflex, accompanied by body and brain weight loss, and eventual death (Figures 1B and S1D–S1F; Video S1).

However, when we further crossed *Gba^{ΔNes}* mice with Mincle-deficient mice, motor dysfunction, and these developmental defects were ameliorated (Figures 1B, S1D, and S1E; Video S1). In addition to this acute genetic model with severe phenotypes, we also utilized a drug-induced model of mild GD.⁸ The administration of a GCase inhibitor, conduritol B epoxide (CBE), to mice caused similar symptoms with delayed onset compared with *Gba^{ΔNes}* mice (Figures 1C and 1D). However, the Mincle-deficient background completely protected motor function (Figures 1C and 1D) and all mice survived (Figure 1E).

We next asked which Mincle-expressing cells were responsible for this severe phenotype. Although peripheral myeloid cells were possible candidates,¹³ no increased infiltration of blood-derived immune cells was detected within the brain in GD models (Figures 1F, 1G, and S1G), presumably because blood-brain barrier (BBB) is relatively maintained in GD as reported.¹⁴ Instead, among neurons and glial cells, microglia expressed a substantial amount of Mincle on their surface in *Gba^{ΔNes}* mice (Figures 1H and S1H). Since neurons can express Mincle,¹⁰ we investigated its contribution in neurons and microglia, by establishing *Clec4e*-flox mice followed by crossing *Nes-cre* or *Cx3cr1-cre* mice (Figures S1I–S1K). Mincle expressed in

Cx3cr1-expressing cells, mainly microglia in brain in this setting, was necessary for drug-induced GD mice to display symptoms (Figures 1I and 1J). This requirement of Mincle in these cells was also confirmed by the genetic models of GD (Figures 1K and S1L; Video S2). β -GlcCer accumulation in the brain was not affected by Mincle deficiency (Figure S2A; Table S1). Taken together, Mincle expressed in microglia is likely responsible for neuropathic symptoms in GD.

Microglia activation contributes to GD

We thus investigated microglial status in whole brain sections. In wild-type (WT) mice, microglia stained with ionized calcium-binding adaptor molecule 1 (Iba1) were small and uniformly scattered throughout the whole brain (Figure 2A). However, in *Gba^{ΔNes}* mice, the size and fluorescence intensity of Iba1 staining was increased (Figures 2A–2C), indicating that microglia were activated in some brain regions of *Gba^{ΔNes}* mice, such as the cortex. Indeed, the Iba1⁺ area in the cortex was significantly increased with age and disease progression (Figure S2B). Furthermore, unbiased analysis of single-cell (sc) mRNA-based clustering revealed that the most distinguishable change between WT and *Gba^{ΔNes}* was the appearance of a cluster characterized by microglia activation markers, *Cst7*, *Spp1*, *Ccl3*, *Igf1*, *Gpnmb*, *Itgax*, and *Lgals3*^{15–21} (Figures 2D, 2E, and S2C).

One of the likely triggers of microglia activation would be β -GlcCer because it accumulates in the brain of *Gba^{ΔNes}* mice and acts as a direct ligand for Mincle.⁹ We thus attempted to visualize the localization of β -GlcCer by mass spectrometry imaging (MSI). As shown in Figure 2F, the distribution of β -GlcCer d18:1/18:0, the most abundant species (Table S1),²² was overlapped with the areas where microglia are activated. Indeed, *in vitro* stimulation of isolated microglia with β -GlcCer induced activation status, such as increased size and inflammatory cytokine production via Mincle (Figures S2D and S2E).

To examine the causal effect of activated microglia on GD symptoms, we used minocycline, a known inhibitor of microglia activation.²³ Minocycline treatment suppressed Iba1⁺ area (Figures S2F and S2G), neurological phenotypes (Figure 2G), and both body and brain weight loss in *Gba^{ΔNes}* mice (Figures S2H and S2I). These results suggest that activated microglia mediates GD symptoms.

Activated microglia phagocytose living neurons leading to neuronal loss

We further investigated the molecular mechanisms behind this relationship by evaluating degenerating neurons stained with Fluoro-Jade C (FJC).²⁴ FJC⁺ degenerating neurons are barely detected in the brain of WT mice but were clearly observed in the cortex, thalamus (TH), and hippocampus (HIP) of *Gba^{ΔNes}* mice (Figure 2A). Upon closer examination of the primary motor cortex (MOp), degenerating neurons were most frequently

(E) UMAP plots show microglia activation score. The cluster with the highest activation score is surrounded by a black line.

(F) β -GlcCer d18:1/18:0 (*m/z*: 718.604) in the coronal brain section from P10 WT and *Gba^{ΔNes}* mice visualized by MSI together with immunohistochemistry (IHC) of Iba1 (red) of P21 WT and *Gba^{ΔNes}* mice.

(G) Percentage of loss of righting reflex of *Gba^{ΔNes}* mice with vehicle (Veh; *n* = 11) or minocycline (Mino; *n* = 7) treatment.

Scale bars: 1.0 mm in (A top) and (F); 500 μ m in (A middle); 100 μ m in (A bottom); and 10 μ m in (B). Data were analyzed by the one-way ANOVA followed by Tukey-Kramer test (C) and the log-rank test (G). **p* < 0.05; ***p* < 0.01; *****p* < 0.0001. See also Figure S2.

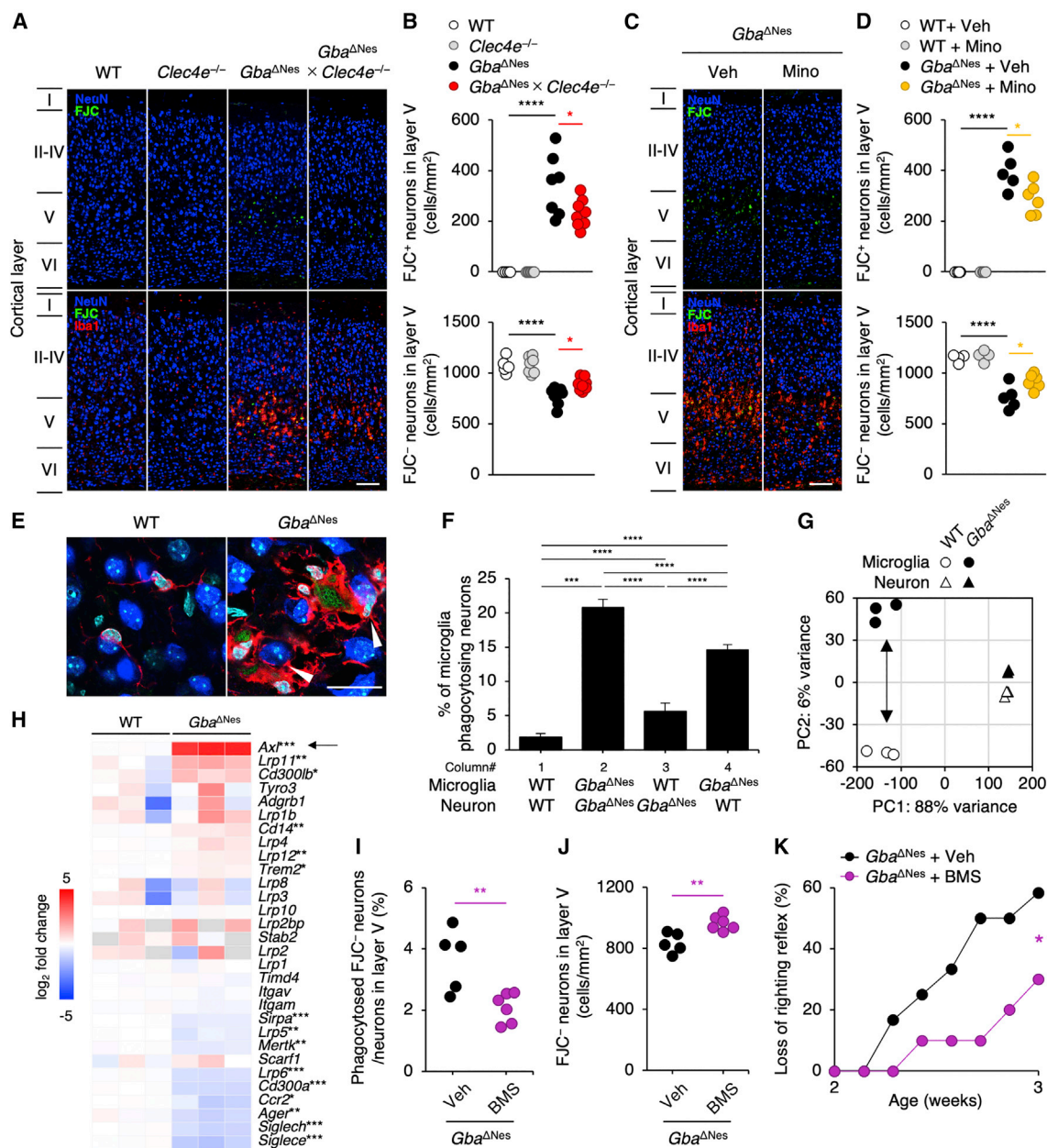


Figure 3. Exacerbation of GD symptoms through phagocytosis of living neurons by activated microglia

(A) Cortical layers of P21 WT, *Clec4e*^{-/-}, *Gba*^{ΔNes}, and *Gba*^{ΔNes} × *Clec4e*^{-/-} mice stained with NeuN (blue), FJC (green), and Iba1 (red).
 (B) Number of FJC⁺ or FJC⁻ neurons in cortical layer V of P21 WT (n = 5), *Clec4e*^{-/-} (n = 7), *Gba*^{ΔNes} (n = 7), and *Gba*^{ΔNes} × *Clec4e*^{-/-} (n = 8) mice.
 (C) Cortical layers of P21 *Gba*^{ΔNes} mice with or without Mino treatment stained with FJC (green), NeuN (blue), and Iba1 (red).
 (D) Number of FJC⁺ or FJC⁻ neurons in cortical layer V of P21 WT and *Gba*^{ΔNes} mice with (WT, n = 5; *Gba*^{ΔNes}, n = 6) or without (WT, n = 4; *Gba*^{ΔNes}, n = 5) Mino treatment.
 (E) Cortical layer V from P21 WT and *Gba*^{ΔNes} mice stained with NeuN (blue), DAPI (cyan), FJC (green), and Iba1 (red). Arrowheads and arrows indicate living neurons (FJC⁺ NeuN⁺) and degenerating neurons (FJC⁺) phagocytosed by microglia.
 (F) *In vitro* phagocytosis assay using microglia and neurons isolated from P21 WT and *Gba*^{ΔNes} mice. Percentage of microglia-phagocytosing neurons (CD11b⁺pHrodo⁺ cells/CD11b⁺ cells) from five technical replicates. Error bars show mean ± standard deviation.
 (G) Principal component analysis plot of RNA-seq data of microglia and neurons from P14 WT and *Gba*^{ΔNes} mice (n = 3 mice per group).
 (H) Heatmap showing genes encoding phagocytic receptors in microglia from P14 *Gba*^{ΔNes} mice compared with P14 WT mice (n = 3 mice per group). Arrow indicates *Axl*.
 (I) Percentage of living neurons phagocytosed by microglia in P21 *Gba*^{ΔNes} mice with (n = 5) or without (n = 6) BMS-777607 treatment.
 (J) Number of FJC⁻ neurons in layer V of P21 *Gba*^{ΔNes} mice with (n = 6) or without (n = 5) BMS-777607 treatment.
 (K) Percentage of loss of righting reflex of *Gba*^{ΔNes} mice with (n = 10) or without (n = 12) BMS-777607 treatment.

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detected in cortical layer V of *Gba*^{ΔNes} mice (Figures 3A, 3B, and S3A), which regulates voluntary motor control.²⁵ Conversely, the number of living neurons (FJC⁺NeuN⁺) was decreased in the same layer of *Gba*^{ΔNes} mice (Figures 3A, 3B, and S3B). This cortical layer V was the exact region where activated microglia accumulated adjacent to degenerating neurons (Figure 3A). Cortical layers were also confirmed with Nissl staining (Figures S3C and S3D). Importantly, constraining microglia activation by deleting Mincle or by minocycline treatment (Figures 2C and S2F) decreased the number of degenerating neurons and increased the number of living neurons in cortical layer V (Figures 3A–3D). Thus, neuronal loss correlates with microglia activation and GD phenotypes.

To investigate the mechanism by which microglia activation leads to neuronal loss, we further magnified cortical layer V. Activated microglia, which show a characteristic amoeboid morphology (Figure 2B),²⁶ engulfed degenerating neurons as expected (Figures 3E, S3E, and S3F). Notably, these microglia also robustly phagocytosed living neurons, which is rarely observed in WT mice (Figures 3E and S3E–S3G; Video S3). This augmented phagocytosis was recapitulated by an *in vitro* co-culture system using microglia and neurons separately isolated from WT and *Gba*^{ΔNes} mice (Figure 3F). The combination of microglia and neurons from WT mice did not show substantial phagocytosis as assessed by pHrodo-labeled neuron uptake (Figure 3F, column 1). However, when we used microglia and neurons from *Gba*^{ΔNes} mice, the proportion of microglia-phagocytosing neurons was increased 10-fold (Figure 3F, column 2). Microglia from *Gba*^{ΔNes} mice mainly contributed to this effect, as WT microglia phagocytosed *Gba*^{ΔNes} neurons less efficiently (Figure 3F, column 3). As the genotype of microglia is identical between WT and *Gba*^{ΔNes} mice, the environment may have explained the difference in phagocytic ability. Indeed, principal component analysis of RNA sequencing (RNA-seq) data revealed that the gene expression signature of microglia from *Gba*^{ΔNes} was distinct from WT mice (Figure 3G). Among phagocytosis-related molecules,^{27–29} the most upregulated gene was *Axl* (Figure 3H), a phosphatidylserine (PtdSer)-dependent phagocytic receptor.³⁰ As *Axl* function can be blocked by BMS-777607, an inhibitor which targets its tyrosine kinase domain,³⁰ *Gba*^{ΔNes} mice were treated with this inhibitor. By this treatment, microglial phagocytosis of FJC⁺ neurons, neuronal loss, and neurological phenotypes were ameliorated (Figures 3I–3K and S3G), suggesting that phagocytosis of living neurons contributed the development of GD symptoms.

In addition to the microglia side, GCase-deficient neurons are also prone to be phagocytosed, as WT neurons were less efficiently engulfed by microglia than GCase-deficient neurons (Figure 3F, column 2 vs. 4 and 3 vs. 1). Consistent with this observation, neurons derived from *Gba*^{ΔNes} mice exposed PtdSer, a target for phagocytic receptors,³¹ on their outer membrane (Figures 4A, 4B, and S4A). In addition to intrinsic stresses caused by lipid accumulation,³² the extrinsic inflammatory environment might also sensitize neurons to expose PtdSer on their surface.²⁸

Several inflammatory cytokines were significantly increased in the brain of GD mice, and some were Mincle-dependent (Figures 4C, S4B, and S4C). Among them, tumor necrosis factor (TNF) was highly induced in microglia through Mincle both *in vitro* and *in vivo* and was exclusively expressed by activated microglia (Figure 4D), as annotated in Figure 2E. Consistent with the fact that TNF induces the necroptosis pathway accompanying PtdSer exposure,³³ neurons from *Gba*^{ΔNes} and CBE-treated mice exhibited a necroptotic signature, as assessed by phosphorylated receptor-interacting protein kinase (RIPK) 1, 3, and mixed lineage kinase domain-like pseudokinase (MLKL) (Figure S4D).^{34,35} Indeed, blockade of this pathway by etanercept (ETN), a soluble TNF receptor,³⁶ and necrostatin-1 (Nec-1), a necroptosis inhibitor,³⁴ diminished phagocytosis of living neurons, improved neuronal loss, and ameliorated neurological phenotypes (Figures 4E–4J). These results suggest that TNF may sensitize neurons to be phagocytosed, which also contributes to GD.

Microglia-mediated pathology is a promising therapeutic target for GD

There are several limitations in analyzing brain from GD patients, such as few number patients and the difficulty in obtaining invasive samples. We therefore analyzed postmortem brains of a neuronopathic GD patient and control donor. GFAP⁺ astrocytes and processes were similarly detected in the cortex of control and GD patients (Figure S5A). Iba1⁺ microglia were detected in the GD cortex and the control donor but enlarged amoeboid microglia were observed in GD (Figures 5A, S5B, and S5C), consistent with the mouse model. Moreover, these activated microglia highly expressed phagocytic markers (Figures S5D and S5E) and were phagocytosing living neurons in the GD cortex (Figures 5B and S5F). Microglia detected in GD also expressed TNF (Figures 5C and S5G), as was observed in mouse models (Figures 4D and S2E).

These results led us to search for applicable therapeutic options by targeting this shared pathway. As several interventions improved the symptoms partially when used as a monotherapy (Figures 2G and 4G), we attempted combinatory use of FDA-approved drugs. Consequently, minocycline and ETN effectively decreased microglia activation, protected neurons, diminished neurological phenotypes, and prevented early death (Figures 5D–5G, S5H, and S5I). Altogether, the present study uncovered a microglia-mediated neuronal loss driven by β-GlcCer and provides an immediately translatable therapeutic option against GD.

DISCUSSION

Previous work on GD has described intrinsic mechanisms leading to neuronal death, for example, through α-synuclein aggregation.^{7,37} Additionally, the involvement of complement and autoantibodies in the GD pathology is reported.³⁸ The microglia-mediated pathway we identified in this study may contribute

Scale bars: 100 μm in (A) and (C) and 20 μm in (E). Each dot represents an individual mouse (B, D, G, I, and J). For each group, three sections from each mouse were analyzed and representative images were shown (A, C, and E). Data are representative of two independent experiments (F). The one-way ANOVA followed by Tukey-Kramer test (B, D, and F), the unpaired two-tailed Student's *t* test (H–J), and the log-rank test (K) were used for statistical analysis. **p* < 0.05; ***p* < 0.01; ****p* < 0.001; *****p* < 0.0001. See also Figure S3.

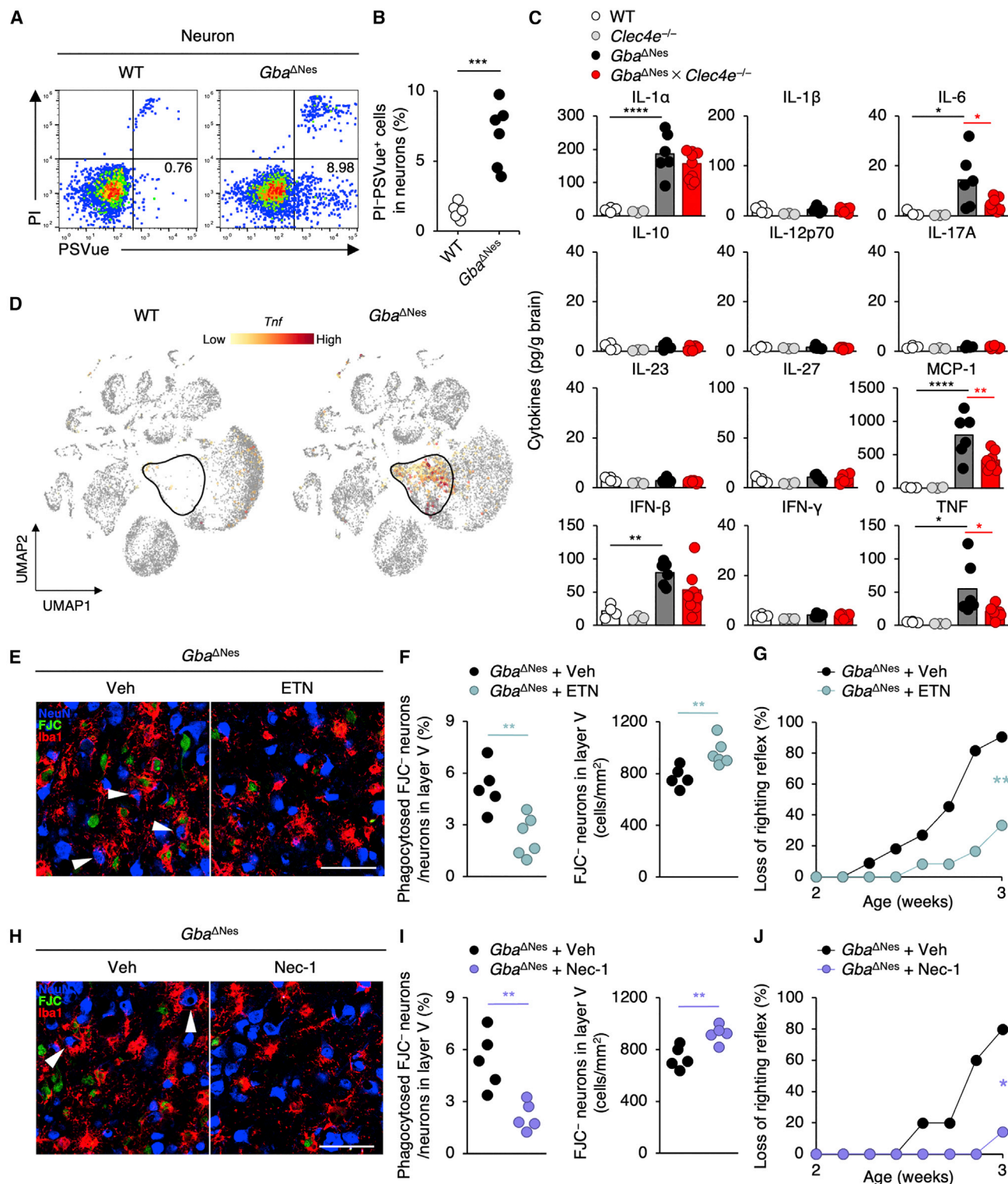


Figure 4. Role of inflammatory environment in GD symptoms

(A) *In vivo* PSVue labeled neurons from P21 WT and $Gba^{\Delta Nes}$ mice.

(B) Percentage of PI⁺PSVue⁺ cells in neurons from P21 WT (n = 5) and $Gba^{\Delta Nes}$ (n = 6) mice.

(C) Inflammatory cytokines in the brain from P21 WT (n = 4), $Clec4e^{-/-}$ (n = 3), $Gba^{\Delta Nes}$ (n = 6), and $Gba^{\Delta Nes} \times Clec4e^{-/-}$ (n = 9) mice. Bars represent the mean for each group.

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to GD independently, or synergistically by promoting such processes, as Mincle is known as an adjuvant receptor.³⁹ Since most of the key symptoms and features observed in mouse GD models used here are shared with GD patients,^{5,8,40–42} the therapies based on this mechanism may be readily translatable.

Mutations in the *GBA* gene, which are detected in up to 20% of Parkinson's disease (PD) patients,⁴³ are associated with earlier onset and worse outcomes such as motor impairments.⁴⁴ Thus, a glucosylceramide synthase (GCS) inhibitor is being investigated in PD patients.⁴⁵ It is possible that microglia-targeted therapies, such as minocycline, may be more effective on stratified PD patients with *GBA* mutation. In addition, as the detrimental phagocytosis described here can also be observed in other neurodegenerative disorders,^{46–49} an inhibitor of this process may be beneficial for these intractable brain diseases.

In addition to Mincle, several C-type lectin receptors (CLRs) recognize lipids and glycoproteins associated with other LSDs, such as Niemann-Pick disease, Sandhoff disease, Mannosidosis, and Sly disease.^{50–53} Thus, the current study points to a previously underappreciated role of the CLR family in LSDs, which warrants further investigation.

The current first-line treatment for neuronopathic GD is ERT which is effective for reducing peripheral symptoms but has limited effect for neurological symptoms.⁶ Additional therapies to address the neurological symptoms of GD are being developed, including a small molecule GCS inhibitor with improved central nervous system bioavailability,⁵⁴ GCase pharmaceutical chaperone,⁵⁵ and virus-based gene therapy,⁵⁶ all have the goal to decrease β -GlcCer accumulation. Since the microglia-targeted therapy proposed here has a different mechanism of action (Figure S5J; Table S1), this could be a next generation therapy applicable together with these β -GlcCer-targeted therapies to treat this fatal disease.

Although we described excessive microglia activation though Mincle exacerbates GD pathogenesis, the physiological role of Mincle in the brain is still a matter of investigation. It may contribute to homeostatic maintenance of neurons through properly regulating microglia activation⁵⁷ by sensing physiological damage.⁵⁸ In line with this idea, during neonatal development, inhibition of microglia leads to brain dysfunction⁵⁹; however, the interventions used here were started after the requirement of microglia activation for neuronal survival and

did not grossly impact brain development (Figures S2H and S2I). As GD is generally diagnosed after 3 months of age,⁶⁰ the advantages of suppressing microglia likely outweigh the possible disadvantages.

In this study, we identified a non-intrinsic pathway of neuronal loss mediated by β -GlcCer-activated microglia. Targeting this pathway by a combination of FDA-approved drugs was effective and could be a clinically feasible therapy for GD in the near future.

Limitations of the study

In this study, we administered ETN via intracerebroventricular (i.c.v.) injection, due to the restricted bioavailability of protein-based drug in GD, in which BBB is maintained.¹⁴ Since i.c.v. is applicable but exceptional for patients, improvement of brain delivery, such as conjugation with antibodies against BBB transporter,⁶¹ may increase the effectiveness of TNF blockers even if administered subcutaneously. Minocycline was originally developed as antibiotic⁶² and the mechanism for how it suppresses microglia is unclear. Further development of derivatives improving microglia specificity⁶³ or brain-targeting would benefit clinical application of this approach.

As phagocytosis is dynamic, static images do not allow us to visualize the whole process. Future development of time-lapse interval imaging would provide a better approach to fully understand where and how phagocytosis contributes to neuron loss in GD.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- RESOURCE AVAILABILITY
 - Lead contact
 - Materials availability
 - Data and code availability
- EXPERIMENTAL MODEL AND SUBJECT DETAILS
 - Mice
 - Human tissue
 - Primary microglia culture

(D) Expression of *Tnfr* in the brain from P14 WT and *Gba*^{ΔNes} mice brain are shown as a heatmap in UMAP plots. The cluster with the highest microglia activation score is surrounded by a black line.

(E) Cortical layer V from P21 *Gba*^{ΔNes} mice with or without etanercept (ETN) treatment stained with NeuN (blue), FJC (green), and Iba1 (red). Arrowheads indicate living neurons (FJC⁺NeuN⁺) phagocytosed by microglia.

(F) Percentage of living neurons phagocytosed by microglia in P21 *Gba*^{ΔNes} mice with (n = 6) or without (n = 5) ETN treatment (left). Number of FJC⁺ neurons in layer V of P21 *Gba*^{ΔNes} mice with (n = 6) or without (n = 5) ETN treatment (right).

(G) Percentage of loss of righting reflex of *Gba*^{ΔNes} mice with (n = 12) or without (n = 11) ETN treatment.

(H) Cortical layer V from P21 *Gba*^{ΔNes} mice with or without necrostatin-1 (Nec-1) treatment stained with NeuN (blue), FJC (green), and Iba1 (red). Arrowheads indicate living neuron (FJC⁺NeuN⁺) phagocytosed by microglia.

(I) Percentage of living neurons phagocytosed by microglia in P21 *Gba*^{ΔNes} mice with or without Nec-1 treatment (n = 5 mice per group) (left). Number of FJC⁺ neurons in layer V of P21 *Gba*^{ΔNes} mice with or without Nec-1 treatment (n = 5 mice per group) (right).

(J) Percentage of loss of righting reflex of *Gba*^{ΔNes} mice with (n = 7) or without (n = 5) Nec-1 treatment.

Scale bars: 50 μ m in (E) and (H). Each dot represents an individual mouse (B, C, F, and I). Data are representative of at least five mice per group (A, E, and H). For each group, three sections per mouse were analyzed and representative images were shown (E and H). The unpaired two-tailed Student's t test (B, F, and I), the one-way ANOVA followed by Tukey-Kramer test (C), and the log-rank test (G and J) were used for statistical analysis. *p < 0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001.

See also Figure S4.

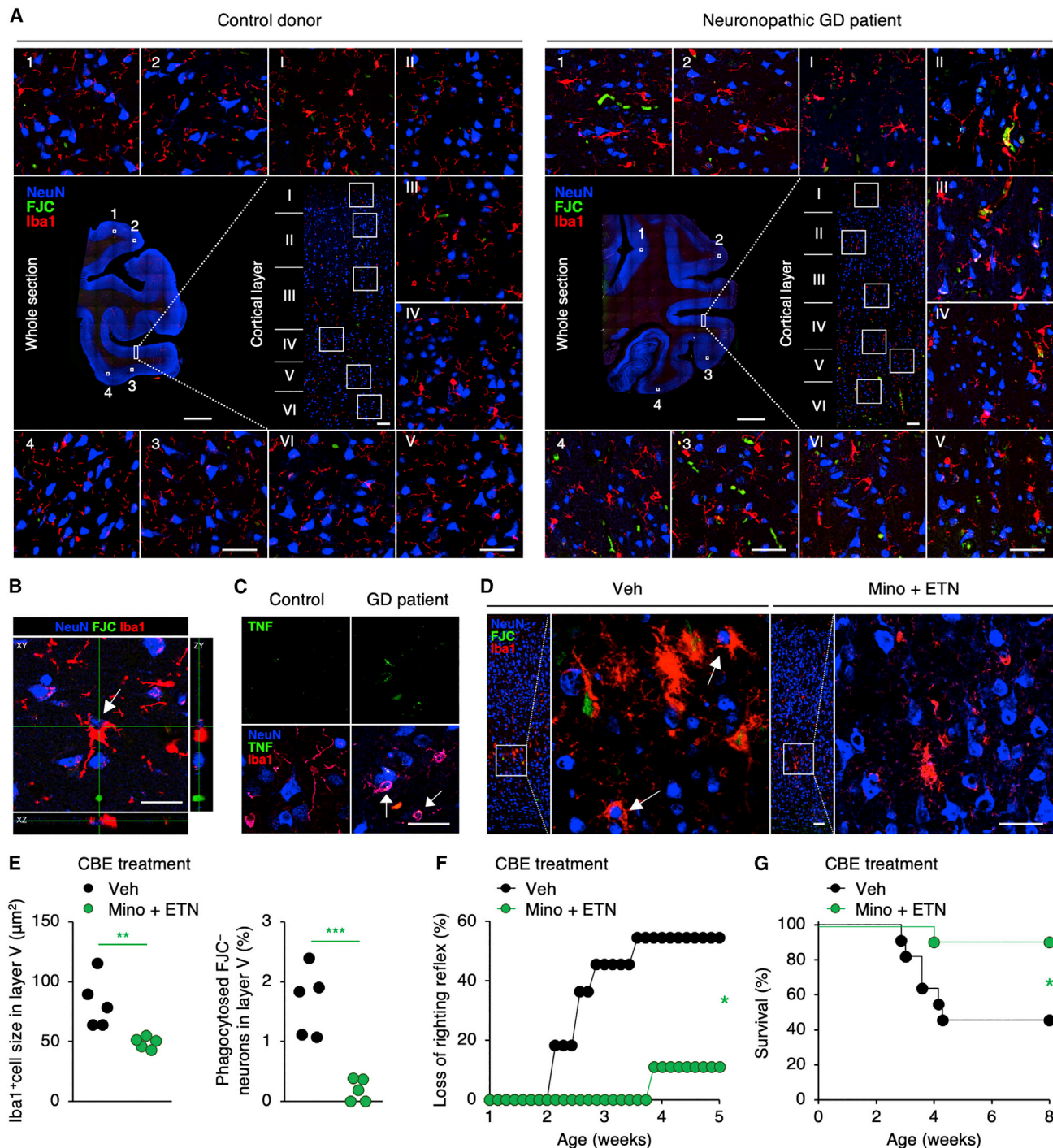


Figure 5. Microglia activation as a target for GD treatment

(A) Control and neuronopathic GD patients stained with NeuN (blue), FJC (green), and Iba1 (red).
 (B) Cortical layer V from a neuronopathic GD patient stained with NeuN (blue), FJC (green), and Iba1 (red) in XY, XZ, and ZY projections. Arrow indicates living neuron (FJC⁺NeuN⁺) phagocytosed by microglia.
 (C) Cortical layer V from control and neuronopathic GD patients stained with NeuN (blue), TNF (green), and Iba1 (red). Arrows indicate TNF⁺ microglia.
 (D) Cortical layer of P21 WT mice with CBE or CBE + Mino + ETN treatment stained with NeuN (blue), FJC (green), and Iba1 (red). Arrows indicate living neurons (FJC⁺NeuN⁺) phagocytosed by microglia.
 (E) Iba1⁺ cell size in cortical layer V (n = 5 mice per group) (left). Percentage of living neurons phagocytosed by microglia in P21 WT mice with CBE or CBE + Mino + ETN treatment (n = 5 mice per group) (right). Each dot represents an individual mouse.
 (F) Percentage of loss of righting reflex of WT mice treated with CBE (n = 11) or CBE + Mino + ETN (n = 9).

(legend continued on next page)

METHOD DETAILS

- SFC/MS/MS analysis
- Drug treatments
- Motor phenotype assessment
- Immunoblots
- Brain tissue dissociation
- Flow cytometry
- Immunohistochemistry (IHC)
- Morphological analysis of microglia
- Quantification of cells
- Single-cell RNA sequencing
- MSI analysis
- Immunocytochemistry
- Magnetic-activated cell sorting
- Cytokine measurement
- Microglial phagocytosis assay
- Bulk RNA-seq
- PtdSer labeling

QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.immuni.2023.01.008>.

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AUTHOR CONTRIBUTIONS

T.S., Y.I., N.T., Z.O., and K.K. did investigation; M.T. provided resources; Y.I., N.T., T.B., J.A., H.T., and A.K. did data curation; S.Y. supervised the research; T.S., C.R.S., and S.Y. wrote the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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(G) Kaplan-Meier survival curve of WT mice treated with CBE (n = 11) or CBE + Mino + ETN (n = 9).

Scale bars: 5 mm in (A whole section); 100 μ m in (A cortical layer); 50 μ m in (A 1–4 and I–VI); and 30 μ m in (B)–(D). Images are representative of one section per patient (A–C). For each group, three sections from three mice were analyzed and representative images were shown (E). The unpaired two-tailed Student's t test (E) and the log-rank test (F and G) were used for statistical analysis. *p < 0.05; **p < 0.01; ***p < 0.001. See also Figure S5.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Mouse anti- β -glucosidase (B-6)	Santa Cruz Biotechnology	Cat# sc-166407; RRID: AB_2109068
Rat anti-Mincle (1B6)	MBL	Cat# D266-3; RRID: AB_10950261
Rabbit anti-RIP (D94C12)	Cell Signaling Technology	Cat# 3493; RRID: AB_2305314
Rabbit anti-RIP3	ProSci Inc.	Cat# 2283; RRID: AB_203256
Rabbit anti-MLKL (D6W1K)	Cell Signaling Technology	Cat# 37705; RRID: AB_2799118
Rabbit anti-Phospho-MLKL (Ser345) (D6E3G)	Cell Signaling Technology	Cat# 37333; RRID: AB_2799112
Rabbit anti- β -actin (13E5)	Cell Signaling Technology	Cat# 4970; RRID: AB_2223172
Sheep anti-mouse IgG HRP	Cytiva	Cat# NA931; RRID: AB_772210
Sheep anti-rat IgG	Cytiva	Cat# NA935; RRID: AB_772207
Goat anti-rabbit IgG	Cell Signaling Technology	Cat# 7074; RRID: AB_2099233
Rat anti-CD16/CD32 (2.4G2)	BD Biosciences	Cat# 553142; RRID: AB_394657
Mouse anti-CD45.2 APC (104)	BD Biosciences	Cat# 558702; RRID: AB_1645215
Rat anti-CD45 Alexa Fluor 700 (30-F11)	BioLegend	Cat# 103127; RRID: AB_493714
Rat anti-CD11b APC (M1/70)	BioLegend	Cat# 101212; RRID: AB_312795
Rat anti-CD11b PE (M1/70)	BioLegend	Cat# 101207; RRID: AB_312790
Rat anti-CD11b Brilliant Violet 510 (M1/70)	BioLegend	Cat# 101245; RRID: AB_2561390
Rat anti-CD11b APC-Cy7 (M1/70)	BioLegend	Cat# 101225; RRID: AB_830641
American hamster anti-CD11c PE (N418)	Invitrogen	Cat# 12-0114-82; RRID: AB_465552
American hamster anti-CD11c PE-Cy7 (N418)	BioLegend	Cat# 117318; RRID: AB_493568
Rat anti-F4/80 Brilliant Violet 421 (BM8)	BioLegend	Cat# 123131; RRID: AB_10901171
Rat anti-F4/80 PE (BM8)	Invitrogen	Cat# 14-4801-82; RRID: AB_467558
Rat anti-Ly-6G APC-Cy7 (1A8)	BioLegend	Cat# 127623; RRID: AB_10645331
Rabbit anti-TMEM119 Alexa Fluor 488 (106-6)	Abcam	Cat# ab225497
Mouse anti-O4 APC (O4)	Miltenyi Biotec	Cat# 130-119-155; RRID: AB_2751644
Rat anti-ACSA-2 APC (IH3-18A3)	Miltenyi Biotec	Cat# 130-117-535; RRID: AB_2727978
Non-neuron cell biotin-antibody cocktail	Miltenyi Biotec	Cat# 130-126-603
Rat anti-Mincle biotin (1B6)	MBL	Cat# D266-6
Rabbit anti-Iba1 (EPR16588)	Abcam	Cat# ab178846; RRID: AB_2636859
Guinea pig anti-IBA1	Synaptic Systems	Cat# 234 004; RRID: AB_2493179
Mouse anti-NeuN (1B7)	Abcam	Cat# ab104224; RRID: AB_10711040
Rat anti-CD68 (FA-11)	Abcam	Cat# ab53444; RRID: AB_869007
Rabbit anti-CD68 (EPR20545)	Abcam	Cat# ab213363; RRID: AB_2801637
Rabbit anti-LAMP1 (EPR21026)	Abcam	Cat# ab208943
Chicken anti-GFAP	Abcam	Cat# ab4674; RRID: AB_304558
Rabbit anti-Axl (C89E7)	Cell Signaling Technology	Cat# 8661; RRID: AB_11217435
Rabbit anti-TNF- α	Cell Signaling Technology	Cat# 3707; RRID: AB_2240625
Goat anti-rabbit IgG H&L Alexa Fluor 405	Abcam	Cat# ab175652; RRID: AB_2687498
Goat anti-rabbit IgG H&L Alexa Fluor 488	Abcam	Cat# ab150077; RRID: AB_2630356
Goat anti-rabbit IgG H&L Alexa Fluor 594	Abcam	Cat# ab150080; RRID: AB_2650602
Goat anti-mouse IgG H&L Alexa Fluor 647	Abcam	Cat# ab150115; RRID: AB_2687948
Goat anti-rat IgG H&L Alexa Fluor 405	Abcam	Cat# ab175671; RRID: AB_2890626
Goat anti-guinea pig IgG H&L Alexa Fluor 594	Abcam	Cat# ab150188; RRID: AB_2905555
Goat anti-chicken IgY H&L Alexa Fluor 594	Abcam	Cat# ab150176; RRID: AB_2716250

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Rabbit anti-Iba1	Wako	Cat# 019-19741; RRID: AB_839504
Goat anti-rabbit HRP labeled polymer	Dako	Cat# K4003; RRID: AB_2630375
Biological samples		
Temporal lobe tissue sample from type 3 GD patient	This paper	N/A
Temporal lobe tissue sample from control donor	This paper	N/A
Chemicals, peptides, and recombinant proteins		
EDTA	Nacalai Tesque	Cat# 15130-95
DNase I	Roche	Cat# 11284932001
Dulbecco's modified Eagle medium	Sigma-Aldrich	Cat# D5796
Penicillin	Sigma-Aldrich	Cat# P3032
Streptomycin	MP Biochemicals	Cat# 194541
Heat-inactivated fetal bovine serum	Sigma-Aldrich	Cat# 175012
Conduritol B epoxide	Calbiochem	Cat# 234599
Minocycline	Sigma-Aldrich	Cat# M9511
Necrostatin-1	Santa Cruz Biotechnology	Cat# sc-200142
Etanercept	Pfizer	N/A
BMS-777607	ChemScene	Cat# CS-0227
Nonidet P-40	Nacalai Tesque	Cat# 23640-94
Aprotinin	Sigma-Aldrich	Cat# A6012
Antipain dihydrochloride	Sigma-Aldrich	Cat# A6191
Chymostatin	Sigma-Aldrich	Cat# C7268
Leupeptin trifluoroacetate salt	Sigma-Aldrich	Cat# L2023
Pepstatin A	Sigma-Aldrich	Cat# P4265
Block Ace	KAC Co. Ltd.	Cat# UKB40
Chemi-Lumi One Super	Nacalai Tesque	Cat# 02230-30
Bovine serum albumin	Nacalai Tesque	Cat# 01860-07
Percoll	Cytiva	Cat# 17089101
Streptavidin PE	BioLegend	Cat# 405203
Propidium iodide	Sigma-Aldrich	Cat# P4170
4%-paraformaldehyde phosphate buffer solution	Nacalai Tesque	Cat# 09154-85
O.C.T. Compound	Sakura Finetek Japan	Cat# 4583
Triton X-100	Nacalai Tesque	Cat# 28229-25
Protein Block	Dako	Cat# X090930-2
DAPI	Invitrogen	Cat# D1306
FluorSave	Calbiochem	Cat# 345789
DAB	Nichirei Biosciences	Cat# 425011
Sodium carboxymethylcellulose	Wako	Cat# 039-01335
pHrodo Green STP Ester	Invitrogen	Cat# P35369
QIAzol Lysis Reagent	Qiagen	Cat# 79306
TruSeq Stranded mRNA Library Prep Kit	Illumina	Cat# RS-122-2101
PSVue 643	Molecular Targeting Technologies	Cat# P-1006
Critical commercial assays		
Pierce™ BCA Protein Assay Kit	Thermo Fisher Scientific	Cat# 23225
Viridis BEH column	Waters	Cat# 186006564
Papain Dissociation System	Worthington Biochemical Corp.	Cat# LK003150
Fluoro Jade C Ready to Dilute Stain Kit	Biosensis	Cat# TR-100-FJT
LEGENDplex™ Mouse Inflammation Panel (13-plex)	BioLegend	Cat# 740150

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
bulk RNA-seq data	This paper	GEO: GSE202711
scRNA-seq data	This paper	GEO: GSE202929
Experimental models: Organisms/strains		
Mouse: WT: C57BL/6J	CLEA Japan, Inc.	C57BL/6JJcl
Mouse: <i>Clec4e</i> ^{-/-} : B6.Mincle ^{-/-}	Yamasaki et al. ⁶⁴	N/A
Mouse: <i>Nes-cre</i> : B6.Cg-Tg(<i>Nes-cre</i>)1Kln/J	The Jackson Laboratory	RRID: IMSR_JAX:003771
Mouse: <i>Cx3cr1-cre</i> : B6J.B6N(Cg)- <i>Cx3cr1</i> ^{tm1.1(cre)Jung/J}	The Jackson Laboratory	RRID: IMSR_JAX:025524
Mouse: CAG-FLPe: B6. Tg(CAG-flpe)36lto	RIKEN	RRID: IMSR_RBRC01834
Mouse: <i>Gba</i> ^{flox/flox} : B6. <i>Gba</i> ^{flox/flox}	This paper	N/A
Mouse: <i>Clec4e</i> ^{flox/flox} : B6. <i>Clec4e</i> ^{flox/flox}	This paper	N/A
Oligonucleotides		
Primer for mouse <i>Gba</i> Forward: CCTCAACTCCTCAAGATCTACTTGCC	Eurofins Scientific	N/A
Primer for mouse <i>Gba</i> Reverse: CAGCTGAGGCTCAGGGCTAAGAG	Eurofins Scientific	N/A
Primer for mouse <i>Mincle</i> Forward: GGTCAGGATGAGGACACAACAA	Eurofins Scientific	N/A
Primer for mouse <i>Mincle</i> Reverse: GATTCGTACTCTCCCTCCATT	Eurofins Scientific	N/A
Software and algorithms		
LabSolution software v5.99 SP2	Shimadzu Corp.	https://www.shimadzu.eu/labsolutions-0
ImageJ	Schneider et al. ⁶⁵	https://imagej.nih.gov/ij/
FlowJo v10.5.2	Tree Star Inc.	https://www.flowjo.com/
BBrowser v2.10.48	BioTuring	https://bioturing.com/bbrowser
Mirion	Paschke et al. ⁶⁶	https://pubs.acs.org/doi/full/10.1007/s13361-013-0667-0
LEGENDplex v8.0	BioLegend	https://www.biolegend.com/en-us/legendplex/software
TopHat v2.0.13	Trapnell et al. ⁶⁷	https://cole-trapnell-lab.github.io/projects/tophat/
Bowtie2 v2.2.3	Langmead and Salzberg ⁶⁸	http://bowtie-bio.sourceforge.net/bowtie2/index.shtml
SAMtools v0.1.19	Li et al. ⁶⁹	http://samtools.sourceforge.net/
Cufflink v2.2.1	Trapnell et al. ⁷⁰	http://cole-trapnell-lab.github.io/cufflinks/
iDEP v0.91	Ge et al. ⁷¹	http://bioinformatics.sdstate.edu/idep/
DESeq2	Love et al. ⁷²	https://bioconductor.org/packages/release/bioc/html/DESeq2.html
GraphPad Prism v9.1.0	GraphPad Software	https://www.graphpad.com/
Other		
Poly-D-lysine-coated T-75 flask	Corning	Cat# 354537
Neuros syringe	Hamilton Company	Cat# 4015-63014
Immobilon-P PVDF membranes	Merck Millipore	Cat# IPVH20200
MAS-coated glass slide	Matsunami Glass Industrial	Cat# MAS-01
Chromium Next GEM Chip G Single Cell Kit	10X Genomics	Cat# PN-1000127
Chromium Next GEM Single Cell 5' Kit v2	10X Genomics	Cat# PN-1000265
MS column	Miltenyi Biotec	Cat# 130-042-201
MACS Separator	Miltenyi Biotec	Cat# 130-042-501
Mouse anti-APC microbeads	Miltenyi Biotec	Cat# 130-090-855

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Mouse anti-biotin microbeads	Miltenyi Biotec	Cat# 130-090-485
TruSeq Stranded mRNA Library Prep Kit	Illumina	Cat# RS-122-2101

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Sho Yamasaki (yamasaki@biken.osaka-u.ac.jp).

Materials availability

Materials generated in this study are available upon request from the lead contact, Sho Yamasaki (yamasaki@biken.osaka-u.ac.jp).

Data and code availability

- The bulk RNA-seq and scRNA-seq data have been deposited at GEO and are publicly available as of the data of publication, Accession numbers are listed in the [key resources table](#). All data reported in this paper will be shared by the [lead contact](#) upon request.
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Mice

Mincle-deficient mice⁶⁴ were backcrossed for at least 23 generations with C57BL/6J (WT) mice, which were obtained from CLEA Japan, Inc. *Nes-cre* and *Cx3cr1-cre* mice were purchased from The Jackson Laboratory. Both male and female mice were used for all experiments. *Gba*-floxed mice and *Clec4e*-floxed mice were generated by gene targeting. Targeting vectors containing a neomycin resistant (Neo^r) cassette and loxP sites were used for homologous recombination in embryonic stem cells of WT mice. The Neo^r cassette of each allele was removed by crossing with CAG-FLPe mice and floxed mice were backcrossed to WT mice at least six times. WT, floxed, and deleted alleles were identified by polymerase chain reaction (PCR) using the following primer pairs: 5'-CCTTCAACTCCTCAAGATCTACTTGCC-3' and 5'-CAGCTGAGGCTCAGGGCTAAGAG-3' for *Gba* and 5'-GGTCAGGATGAGGACACAACAA-3' and 5'-GATTCGTAATCTCCCTCCATT-3' for *Clec4e*. All mice were maintained in laminar-flow cages with filtered air under specific pathogen-free conditions and given standard laboratory food and water ad libitum. All animal procedures were approved by the Committee of Ethics on Animal Experiments at the Research Institute for Microbial Diseases, Osaka University, Japan.

Human tissue

The present study was approved by the ethics committees of the Research Institute for Microbial Disease at Osaka University (RIMD-2021-10) and National Center of Neurology and Psychiatry (A2022-006). Written informed consents were obtained from the patients. Formalin-fixed temporal lobe tissue samples from a 24-year-old female patient with type 3 GD and a 25-year-old male control donor without neurological disease were obtained from the National Center of Neurology and Psychiatry. The GD patient had severe, progressive symptoms including mental retardation, loss of spontaneous language, progressive myoclonus epilepsy, generalized tonic-clonic seizures, ataxia, difficulty swallowing, and eye movement abnormality.

Primary microglia culture

Primary microglia cultures were prepared as previously described, with some modifications.⁷³ Briefly, minced cerebral cortices from male and female WT mice at P0–2 were dissociated with 5 ml of PBS containing EDTA (1 μ M; Nacalai Tesque, 15130-95) on shaker for 10 min at 180 rpm and treated with DNase I (1 μ g/mL; Roche, 11284932001) for 30 s at 37°C. After centrifugation at 400 $\times$ g for 3 min at 4°C, the dissociated cells were suspended in Dulbecco's modified Eagle medium (DMEM, Sigma-Aldrich, D5796) supplemented with 0.1% penicillin (Sigma-Aldrich, P3032)/streptomycin (MP Biochemicals, 194541) and 10% heat-inactivated fetal bovine serum (Sigma-Aldrich, 175012). Next, the cells were seeded into poly-D-lysine-coated T-75 flasks (Corning, 354537) and cultured in a 5% CO₂ incubator at 37°C for 2–3 weeks. Finally, the microglia were harvested by shaking the flasks for 2 h at 180 rpm.

METHOD DETAILS

SFC/MS/MS analysis

Brain samples from postnatal day 10 (P10), P14, and P21 mice were prepared for lipid extraction using the Bligh and Dyer method.⁷⁴ Briefly, the lipids were extracted from frozen and crushed brain tissue with 990 μ L of methanol and 10 μ L of internal standard solution containing β -GalCer d18:1(d₇)/13:0 (100 pmol), β -GlcCer d18:1(d₅)/18:1 (100 pmol), β -GalSPB d18:1 (d₅) (100 pmol), and β -GlcSPB d18:1 (d₅) (100 pmol). The samples were vigorously mixed for 1 min and sonicated for 5 min. To precipitate protein, the methanol extracts were incubated on ice for 5 min. The extracts were then centrifuged at 16,000 $\times$ g for 5 min at 4°C, and the resultant supernatant was collected. Protein concentrations in the pellet were determined using a PierceTM BCA Protein Assay Kit (Thermo Fisher Scientific, 23225). The collected supernatant (400 μ L) was mixed with 400 μ L of chloroform, and 320 μ L of water, and then centrifuged at 16,000 $\times$ g for 5 min at 4°C. The organic layer (280 μ L) was dried under nitrogen and stored at -80°C until analysis.

SFC/MS/MS analysis was performed as previously described⁷⁵ using a Nexera UC system (Shimadzu Corp.) under the following operating conditions: column, Viridis BEH column (3.0 $\times$ 100 mm, 1.7 μ m, Waters, 186006564); column temperature, 40°C; mobile phase A, supercritical carbon dioxide; mobile phase B, methanol/water (95:5, v/v) with 0.1% (w/v) ammonium acetate; flow rate of mobile phase, 1.0 mL/min; and flow rate of make-up pump, 0.1 mL/min. The gradient conditions were as follows: 1–50% B, 0–17 min; 50% B, 17–26 min; and 1% B, 26–30 min. Triple quadrupole MS was performed using an LCMS-8060 system (Shimadzu Corp.). The operating conditions were as follows: polarity, positive ionization; electrospray voltage, 4.0 kV; desolvation line temperature, 250°C; heat block temperature, 400°C; nebulizing gas flow rate, 3.0 L/min; drying gas flow rate, 10.0 L/min; collision-induced dissociation gas pressure, 0.23 MPa; and detector voltage, 2.16 kV. The multiple reaction monitoring parameters per one time period were 2 ms dwell time and pause time, respectively. Data processing was performed using LabSolution software v5.99 SP2 (Shimadzu Corp.). All data were presented in β -anomeric form because mammals cannot synthesize α -anomeric sugars.⁷⁶

Drug treatments

Conduritol B epoxide (CBE; 12.5 mg/kg in PBS; Calbiochem, 234599) was administered daily by intraperitoneal injection from P7. Minocycline (100 mg/kg in PBS, Sigma-Aldrich, M9511), and necrostatin-1 (10 mg/kg in PBS; Santa Cruz Biotechnology, sc-200142) were administered daily by intraperitoneal injection from P10. Etanercept (5 mg/kg in PBS, 2 μ L; Pfizer) and BMS-777607 (10 mg/mL in dimethyl sulfoxide, 1 μ L; ChemScene, CS-0227) were administered every other day by i.c.v. injection using a 33-gauge Neuros syringe (Hamilton Company, 4015-63014) from P10.

Motor phenotype assessment

Locomotor activity was assessed at 4 weeks of age, using the open field test, as previously described.⁷⁷ Briefly, mice were placed in an open field area (25 $\times$ 25 cm²), and exploratory behavior was videotaped for 10 min. After thresholding the videos to create binary images using ImageJ software,⁶⁵ the MTrack2 plugin (<https://imagej.net/plugins/mtrack2>) was used to track the movement of the mice and produce a trace image of the path. Staring at P7, the righting reflex was measured as the time taken for the mice to return to an upright position after being placed on its back. Loss of righting reflex was defined as the inability to right itself within 60 s.⁷⁸

Immunoblots

Immunoblots were performed as previously described.⁷⁹ Briefly, after intracardiac perfusion with PBS, whole brains or isolated neurons were sonicated and suspended in lysis buffer containing Nonidet P-40 (1%; Nacalai Tesque, 23640-94), phenylmethylsulfonyl fluoride (1 mM; Nacalai Tesque, 27327-52), protease inhibitor cocktail containing aprotinin (10 μ g/mL; Sigma-Aldrich, A6012), antipain dihydrochloride (12.5 μ g/mL; Sigma-Aldrich, A6191), chymostatin (12.5 μ g/mL; Sigma-Aldrich, C7268), leupeptin trifluoroacetate salt (50 μ g/mL; Sigma-Aldrich, L2023), and pepstatin A (25 μ g/mL; Sigma-Aldrich, P4265). After centrifugation at 20,000 $\times$ g for 60 min at 4°C, the protein in the supernatants was quantified using a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific). The lysates were mixed with sodium dodecyl sulfate (SDS) sample buffer and boiled for 5 min at 95°C. Protein lysates were separated by 10% SDS-polyacrylamide gel electrophoresis (PAGE) and transferred to Immobilon-P PVDF membranes (Merck Millipore, IPVH20200) which were blocked with Block Ace (KAC Co. Ltd., UKB40) and were incubated overnight at 4°C with the following primary antibodies: mouse anti- β -glucosidase (1:100, B-6; Santa Cruz Biotechnology, sc-166407), rat anti-Mincle (1:100, 1B6; MBL, D266-3), rabbit anti-RIP antibody (1:1,000, D94C12; Cell Signaling Technology, 3493), rabbit anti-RIP3 (1:1,000; ProSci Inc., 2283), rabbit anti-MLKL (1:1000, D6W1K; Cell Signaling Technology, 37705), rabbit anti-phospho-MLKL (1:1000, D6E3G; Cell Signaling Technology, 37333), or rabbit anti- β -actin (1:3,000; 13E5, Cell Signaling Technology, 4970). Then, membranes were incubated for 40 min at room temperature (RT) with the following secondary antibodies: sheep anti-mouse IgG HRP (1:10,000; Cytiva, NA931), sheep anti-rat IgG (1:10,000; Cytiva, NA935), or goat anti-rabbit IgG (1:4,000; Cell Signaling Technology, 7074). Immunoreactive bands were detected with Chemi-Lumi One Super (Nacalai Tesque, 02230-30) using a LAS 1000 luminescent image analyzer (FUJIFILM).

Brain tissue dissociation

Single-cell suspensions of brain tissues were prepared as previously described.⁸⁰ Briefly, mice were sacrificed by CO₂ asphyxiation and then intracardially perfused with PBS. Whole brains were minced with scissors and enzymatically dissociated with Papain dissociation system (Worthington Biochemical Corp., LK003150) containing papain (20 U/mL), L-cysteine (1 mM), DNase I (400 U/mL), and

EDTA (0.5 mM) in Earle's Balanced Salt Solution. Prior to use, papain was equilibrated for 30 min at 37°C with 5% CO₂. After the treatment, the tissues were washed with papain inhibitor buffer containing bovine serum albumin (BSA; 1 mg/mL) and ovomucoid (1 mg/mL) and then mechanically dissociated by gentle pipetting. Finally, the dissociated cells were resuspended in 20% (v/v) Percoll (Cytiva, 17089101) in PBS and centrifuged at 400 × g for 25 min at 4°C to remove the myelin and debris.

Flow cytometry

Single cell pellets were incubated with anti-CD16/CD32 (1:100, 2.4G2; BD Biosciences, 553142) for 20 min at 4°C. Then cells were stained for 20 min at 4°C with anti-CD45.2 APC (1:40, 104; BD Biosciences, 558702), anti-CD45 Alexa Fluor 700 (1:50, 30-F11; BioLegend, 103127), anti-CD11b Brilliant Violet 510 (1:200, M1/70; BioLegend, 101245), anti-CD11b APC-Cy7 (1:200, M1/70; BioLegend, 101225), anti-CD11c PE (1:100, N418; Invitrogen, 12-0114-82), anti-CD11c PE-Cy7 (1:100, N418; BioLegend, 117318), anti-F4/80 PE (1:50, BM8; Invitrogen, 14-4801-82), anti-F4/80 Brilliant Violet 421 (1:50, BM8; BioLegend, 123131), anti-Ly-6G APC-Cy7 (1:200, 1A8; BioLegend, 127623), anti-TMEM119 Alexa Fluor 488 (1:200, 106-6; Abcam, ab225497), anti-O4 APC (1:10, O4; Miltenyi Biotec, 130-119-155), anti-ACSA-2 APC (1:10, IH3-18A3; Miltenyi Biotec, 130-117-535), non-neuron cell biotin-antibody cocktail (1:10; Miltenyi Biotec, 130-126-603), anti-Mincle biotin (1:500, 1B6; MBL, D266-6), and/or streptavidin PE (1:100; BioLegend, 405203). Propidium iodide (PI; Sigma-Aldrich, P4170) was added to label dead cells, which were analyzed using Gallios flow cytometer (Beckman-Coulter). The resultant data were evaluated using FlowJo v10.5.2 software (Tree Star Inc.).

Immunohistochemistry (IHC)

After intracardiac perfusion with PBS, whole brains were fixed in 4% (w/v) paraformaldehyde (PFA; Nacalai Tesque, 09154-85) in PBS overnight. For fluorescent imaging, brains were transferred to 30% (w/v) sucrose (Nacalai Tesque, 30404-45) in PBS until the tissues sank. Then, the cryoprotected tissues were embedded in O.C.T. Compound (Sakura Finetek Japan, 4583), and 12 μm thick sections were cut with a Cryostar NX50 cryostat (Thermo Fisher Scientific) and mounted on MAS-coated glass slides (Matsunami Glass Industrial, MAS-01). The brain tissue sections were permeabilized with 0.1 % Triton X-100 (Nacalai Tesque, 28229-25) and blocked with Protein Block (Dako, X090930-2) for 10 min at RT. Then, the sections were incubated overnight at 4°C with the following antibodies diluted with 1% (w/v) BSA in PBS: rabbit anti-Iba1 (1:1,000, EPR16588; Abcam, ab178846), guinea pig anti-IBA1 (1:500; Synaptic Systems, 234 004), mouse anti-NeuN (1:1,000, 1B7; Abcam, ab104224), rat-anti CD68 (1:500, FA-11; Abcam, ab53444), and/or rabbit anti-LAMP1 (1:200, EPR21026; Abcam, ab208943). Next, the sections were incubated for 2 h at 37°C with the following secondary antibodies diluted with 1% (w/v) BSA in PBS: goat anti-rabbit IgG H&L Alexa Fluor 405 (1:500; Abcam, ab175652), goat anti-rabbit IgG H&L Alexa Fluor 594 (1:500; Abcam, ab150080), goat anti-mouse IgG H&L Alexa Fluor 647 (1:500; Abcam, ab150115), goat anti-rat IgG H&L Alexa Fluor 405 (1:500; Abcam, ab175671), and/or goat anti-guinea pig IgG H&L Alexa Fluor 594 (1:500; Abcam, ab175678). Additionally, FJC staining was performed according to the manufacturer's instructions with some modifications. Briefly, the sections were incubated for 10 min at RT with 0.0001% (v/v) FJC working solution (Biosensis, TR-100-FJT), counterstained with DAPI (3 μg/mL, Invitrogen, D1306), and mounted in FluorSave (Calbiochem, 345789). The sections were visualized with an FV3000 confocal laser scanning microscope (Olympus).

For diaminobenzidine (DAB) staining, the brains were embedded in paraffin, and sections were mounted on the glass slides. After deparaffinization, the sections were washed with Tris-buffered saline (TBS; 0.01M, pH 6.0) for 9 min at 90°C. Endogenous peroxidases were quenched using 0.3% hydrogen peroxidase in methanol for 30 min at RT. Residual enzymatic activity was removed by washing with TBS with Tween (TBS-T), and nonspecific antigen binding was blocked by preincubation with 10% (w/v) normal goat serum in PBS for 30 min at RT. Then, the sections were incubated with rabbit anti-Iba1 antibody (1:250; Wako, 019-19741) for 60 min at RT. After extensive washing with TBS-T, the sections were incubated with HRP-labeled polymer anti-rabbit antibody (Dako, K4003) for 30 min at RT. Immunostaining was detected with DAB (Nichirei Biosciences, 425011) with Nissl counterstaining. Images were acquired using a VS200 slide scanner (Olympus).

For human brain staining, postmortem brains were embedded in paraffin, and sections were mounted on the glass slides. The sections were deparaffinized and antigen retrieval was performed with 10 mM sodium citrate buffer (pH 6.0) for 20 min at 98°C. After cooling, the brain tissue sections were permeabilized with 0.1 % Triton X-100 and blocked with Protein Block for 10 min at RT. Then, the sections were incubated overnight at 4°C with rabbit anti-Iba1 (1:1,000, EPR16588), guinea pig anti-IBA1 (1:500; Synaptic Systems, 234 004), mouse anti-NeuN (1:1,000, 1B7), chicken anti-GFAP (1:1000, Abcam, ab4674), rabbit anti-CD68 (1:1000, EPR20545; Abcam, ab213363), rabbit anti-Axl (1:500, C89E7; Cell Signaling Technology, 8661), and/or rabbit anti-TNF (1:500; Cell Signaling Technology, 3707) diluted with 1% (w/v) BSA in PBS. Next, the sections were incubated for 2 h at 37°C with goat anti-rabbit IgG H&L Alexa Fluor 405 (1:500; Abcam, ab175652), goat anti-rabbit IgG H&L Alexa Fluor 488 (1:500; Abcam, ab150077), goat anti-rabbit IgG H&L Alexa Fluor 594 (1:500; Abcam, ab150080), goat anti-mouse IgG H&L Alexa Fluor 647 (1:500; Abcam, ab150115), goat anti-guinea pig IgG H&L Alexa Fluor 594 (1:500; Abcam, ab175678), and/or goat anti-chicken IgY H&L (1:500; Abcam, ab150176) diluted with 1% (w/v) BSA in PBS. Then, the sections were incubated for 10 min at RT with 0.0001% (v/v) FJC working solution and mounted in FluorSave. The sections were visualized with an FV3000 confocal laser scanning microscope.

Morphological analysis of microglia

Three microglia were randomly selected from the brain of each mouse for morphological analysis as previously described.⁸¹ Briefly, Z-stack confocal microglia images from 50 μm thick, Iba1-stained cerebral cortex sections were acquired at 1 μm intervals using an

FV3000 confocal laser scanning microscope. The consecutive images were converted to a maximum intensity projection image using ImageJ. Then, each image was binarized by applying the automatic threshold, and the Analyze Particle function in ImageJ was used to calculate the size of the microglia body. The process length was calculated by skeletonizing the binary images using the Skeletonize3D plugin (<https://imagej.net/plugins/skeletonize3d>), and the total length of the processes was subsequently calculated by Analyze Skeleton plugin (<https://imagej.net/plugins/analyze-skeleton/>) for ImageJ.

Quantification of cells

Iba1⁺, NeuN⁺, and FJC⁺ cells were quantified using three images for each mouse from stereotactically-identified regions of the motor cortex that were binarized by applying the automatic threshold and calculated using the Analyze Particle function for ImageJ. Phagocytosis of neurons by microglia was analyzed using Z-stack images acquired at 0.4 μ m intervals that were visualized using the Orthogonal View or 3D Viewer plugin (<https://imagej.net/plugins/3d-viewer/>). Phagocytosis was defined as the engulfment of more than 50% of the target neuron.⁸²

Iba1⁺, Iba1⁺FJC⁺NeuN⁺, and Iba1⁺TNF⁺NeuN⁺ cells were quantified by counting five random 0.3125 $\times$ 0.3125 mm images in cortical layer V from control and GD patients.

Single-cell RNA sequencing

ScRNA-seq was performed as previously described.⁸³ Briefly, a single-cell suspension from the brain of P14 mice containing approximately 1×10^4 cells was loaded into a Chromium Next GEM Chip G Single Cell Kit (10x Genomics, PN-1000127), and a Chromium Controller was used for barcoding and cDNA synthesis. A Chromium Next GEM Single Cell 5' Library & Gel Bead Kit v2 (10x Genomics, PN-1000265) was used to amplify the cDNA and construct the cDNA libraries which were sequenced on Illumina NovaSeq 6000 System in paired-end mode (read 1, 28 bp; read 2, 91 bp). Data analysis and visualization were processed by BBrowser v2.10.48 software (BioTuring). Briefly, cells that were detected with a mitochondrial content > 10%, < 200 genes, or > 4,000 genes were considered dying cells, empty droplets, and doublets, respectively, and were excluded. Differential cell types were identified by uniform manifold approximation and projection (UMAP) based on the differentially expressed genes (DEGs) of each cluster. Microglia activation signature scores were generated based on reported microglia activation genes (*Cst7*, *Spp1*, *Ccl3*, *Igf1*, *Gpnmb*, *Itgax*, and *Lgals3*)^{15–21} and shown as a heatmap in UMAP plot using BBrowser.⁸⁴

MSI analysis

Matrix-assisted laser desorption/ionization (MALDI)-MSI was performed as previously described.⁸⁵ Briefly, after intracardiac perfusion with PBS, mouse brains were immediately embedded in 2% sodium carboxymethylcellulose (Wako, 039-01335), cryosectioned at a thickness of 10 μ m, and mounted on MAS-coated glass slides. Then, the brain sections were washed for 10 s with an aqueous buffer containing 50 nM ammonium formate. After drying, the sections were sprayed with para-nitroaniline (p-NA) in 80% ethanol containing 150 mM ammonium formate using a SunCollect sprayer (SunChrom). The first 10 layers were sprayed with 1.5 mg/mL of p-NA, and the next 25 layers were sprayed with 10 mg/mL of p-NA. MALDI-MSI was performed using a Q Exactive Orbital Mass Spectrometer (Thermo Fisher Scientific) connected to an AP-SMALDI⁵ imaging system (TransMIT) in full-scan mode with a mass range of 600–1,000 *m/z* at a mass resolution of 140,000. All ion images were visualized using Mirion software,⁶⁶ with a mass tolerance of ± 10 ppm.

Immunocytochemistry

Primary microglia were fixed with 4% (w/v) PFA in PBS for 15 min at RT, permeabilized with 0.1 % Triton X-100, and blocked with 3% (w/v) BSA for 1 h at RT. Then, the cells were incubated overnight at 4°C with rabbit anti-Iba1 (1:1,000, EPR16588; Abcam) diluted with 1% (w/v) BSA in PBS, followed by goat anti-rabbit IgG H&L Alexa Fluor 594 (1:500; Abcam, ab150080) diluted with 1% (w/v) BSA in PBS for 2 h at 37°C. Finally, the cells were visualized with a BZ-9000 fluorescence microscope (Keyence).

Magnetic-activated cell sorting

Isolated microglia and neurons were prepared as previously described, with some modifications.⁸⁰ Briefly, single-cell suspensions of brain tissues were incubated with anti-mouse CD16/CD32 (1:100, 2.4G2) for 20 min at 4°C. Then, the cells were labeled with anti-CD11b APC (1:200, M1/70; BioLegend, 101212) for 10 min at 4°C and incubated with anti-APC microbeads (1:5; Miltenyi Biotec, 130-090-855) for 10 min at 4°C. Next, the cells were loaded into an MS column (Miltenyi Biotec, 130-042-201) and placed in a MACS Separator (Miltenyi Biotec, 130-042-501). The CD11b-labeled cell fraction was collected as microglia. The non-labeled (negative) cell fraction was further incubated with a non-neuron cell biotin-antibody cocktail for 5 min at 4°C and labeled with anti-biotin microbeads (Miltenyi Biotec, 130-090-485) for 10 min at 4°C. Then, the cells were loaded into an MS column and placed in a MACS Separator. The non-labeled flow-through cell fraction was collected as neurons.

Cytokine measurement

Interleukin (IL)-1 α , IL-1 β , IL-6, IL-10, IL-12p70, IL-17A, IL-23, IL-27, monocyte chemoattractant protein (MCP)-1, interferon (IFN)- β , IFN- γ , and TNF in the supernatants of primary microglia stimulation assays and brain tissue lysates were quantified using the LEGENDplex Mouse Inflammation Panel (13-plex) (BioLegend, 740150) according to the manufacturer's instructions. Following analysis using a Gallios flow cytometer, the resultant data were analyzed with LEGENDplex v8.0 software (BioLegend).

Microglial phagocytosis assay

Microglia from P21 WT and *Gba*^{ΔNes} mice were seeded into 96-well plates at a density of 3×10^4 cells/wells for 24 h. Isolated neurons from P21 WT and *Gba*^{ΔNes} mice were labeled with pHrodo Green STP Ester (Invitrogen, P35369) according to the manufacturer's instructions and cultured with microglia at a density of 3×10^4 cells/well for 3 h. After the treatment, the cells were labeled with anti-CD11b PE antibody (1:200, M1/70; BioLegend, 101207) and analyzed using Gallios flow cytometer. The resultant data were evaluated with FlowJo software.

Bulk RNA-seq

Total RNA was extracted from magnetically sorted microglia and neurons of P14 WT, *Gba*^{ΔNes}, and *Gba*^{ΔNes} × *Clec4e*^{-/-} mice using QIAzol Lysis Reagent (Qiagen, 79306) according to the manufacturer's instructions. The cDNA library was prepared by using a TruSeq Stranded mRNA Library Prep Kit (Illumina, RS-122-2101). Sequencing was performed using an Illumina HiSeq 2500 platform in 75-base single-end mode. Base calling was performed using CASAVA v1.8.2 software (Illumina). Sequenced reads were mapped to the reference mouse genome sequence (mm10) using TopHat v2.0.13⁶⁷ in combination with Bowtie2 v2.2.3⁶⁸ and SAMtools v0.1.19.⁶⁹ The fragments per kilobase of exon per million mapped fragments (FPKMs) were calculated using Cufflink v2.2.1 software.⁷⁰ The read count data were imported into iDEP v0.91,⁷¹ and principal component analysis were performed. DEGs were identified by DESeq2.⁷²

PtdSer labeling

In vivo PtdSer labeling was performed as previously described.⁸⁶ Briefly, 1 μ L of PSVue 643 (1 mM; Molecular Targeting Technologies, P-1006) was injected into the right lateral ventricle using a 33-gauge Neuros syringe. Then, the mice were sacrificed by CO₂ asphyxiation 24 h after injection and intracardially perfused with PBS. The right hemisphere was used for fluorescent imaging. Brain cells from the left cerebrum were stained using a non-neuron biotin-antibody cocktail and streptavidin PE. Then, they were resuspended in PI and analyzed on a Gallios flow cytometer. The resultant data were evaluated using FlowJo software.

QUANTIFICATION AND STATISTICAL ANALYSIS

Data were analyzed using GraphPad Prism v9.1.0 software (GraphPad Software). The unpaired two-tailed Student *t*-test was used to compare the means between two groups. One-way analysis of variance (ANOVA) followed by Tukey–Kramer test was used for multiple group comparisons. Two-way repeated measures ANOVA followed by Fisher's least significant difference (LSD) test was used to compare the mean of two independent repeated variables based on repeated observations. The log-rank test was used to compare the event-time distribution between two groups. *p*-values < 0.05 were considered statistically significant.

Supplemental information

**Direct activation of microglia
by β -glucosylceramide causes phagocytosis
of neurons that exacerbates Gaucher disease**

Takashi Shimizu, Charles R. Schutt, Yoshihiro Izumi, Noriyuki Tomiyasu, Zakaria Omahdi, Kuniyuki Kano, Hyota Takamatsu, Junken Aoki, Takeshi Bamba, Atsushi Kumanogoh, Masaki Takao, and Sho Yamasaki

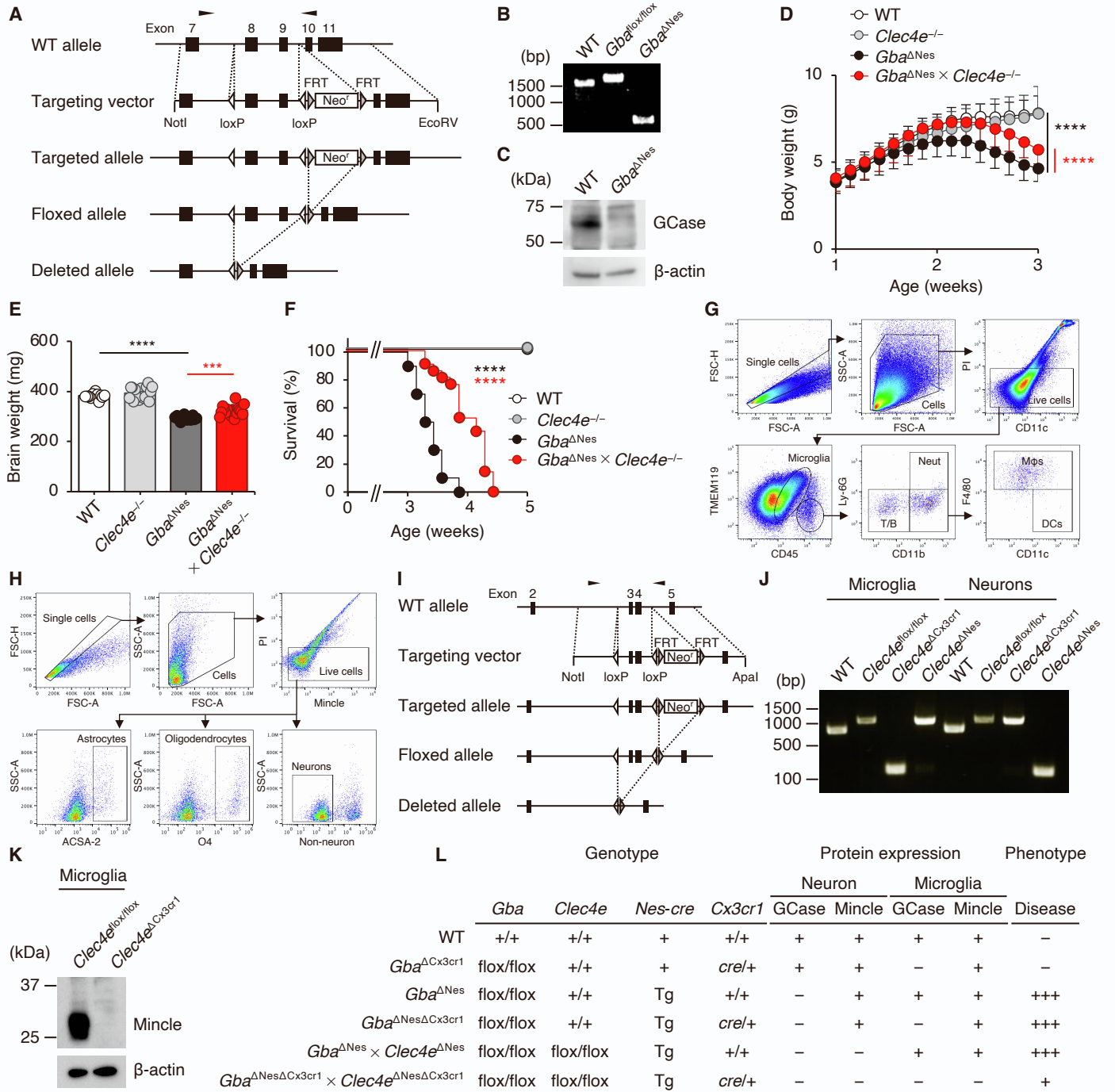


Figure S1. Characterization of mouse GD models (Related to Figure 1)

(A) Schematic view of the generation of *Gba*^{ΔNes} mice. Black bars represent *Gba* exons, open triangles represent loxP sites, gray triangles represent flippase recognition target (FRT) sites, and arrow heads denote primer location. (B) PCR products showing removal of the exons 8 and 9 of *Gba* in the brain of *Gba*^{ΔNes} mice. (C) Western blot analysis of GCase expression in the brain of P21 WT and *Gba*^{ΔNes} mice. (D) Body weight chart for WT, *Clec4e*^{-/-}, *Gba*^{ΔNes}, and *Gba*^{ΔNes} × *Clec4e*^{-/-} mice (n = 15 mice per group). Error bars indicate means ± standard deviation. (E) Brain weight of P21 WT (n = 11), *Clec4e*^{-/-} (n = 17), *Gba*^{ΔNes} (n = 12), and *Gba*^{ΔNes} × *Clec4e*^{-/-} (n = 18) mice. Bars indicate the mean for each group and each dot represents an individual mouse. (F) Kaplan-Meier survival curve of WT (n = 10), *Clec4e*^{-/-} (n = 10), *Gba*^{ΔNes} (n = 10) and *Gba*^{ΔNes} × *Clec4e*^{-/-} mice (n = 21). (G) Representative FACS plots showing the gates for microglia, T and B cells, neutrophils, macrophages, and dendritic cells. (H) Representative FACS plots showing the gates for astrocytes, oligodendrocytes, and neurons. (I) Schematic view of the generation of *Clec4e*^{ΔNes} mice. Black bars represent *Clec4e* exons, open triangles represent loxP sites, gray triangles represent FRT sites, and arrow heads denote primer location. (J) PCR products showing removal of the exons 3 and 4 of *Clec4e* in the microglia and neurons from *Clec4e*^{ΔCx3cr1} and *Clec4e*^{ΔNes} mice. (K) Western blot analysis of Mincle expression in the primary microglia from *Clec4e*^{flox/flox} and *Clec4e*^{ΔCx3cr1} mice stimulated with lipopolysaccharide (LPS; 1 μg/mL) overnight. (L) Schematic view of generation of “double cre × double flox” genetic model. Deficiency of GCase in microglia (WT vs. *Gba*^{flox/flox} × *Cx3cr1*-cre; *Gba*^{ΔCx3cr1} or *Gba*^{ΔNes} vs. *Gba*^{ΔNes} × *Cx3cr1*-cre; *Gba*^{ΔNes} × *Cx3cr1) and Mincle in neuron (*Gba*^{ΔNes} vs. *Gba*^{ΔNes} × *Clec4e*^{flox/flox}; *Gba*^{ΔNes} × *Clec4e*^{ΔNes}) did not change disease. Deficiency of Mincle in microglia (*Gba*^{ΔNes} vs. *Gba*^{ΔNes} × *Cx3cr1 × *Clec4e*^{ΔNes} × *Cx3cr1) ameliorated disease. Data are representative of two (B, C, J, and K) or at least three independent experiments (G and H). The two-way repeated measures ANOVA followed by Fisher's LSD test (D), the one-way ANOVA followed by Tukey-Kramer test (E) and the log-rank test (F) were used for statistical analysis. ***p < 0.001; ****p < 0.0001.***

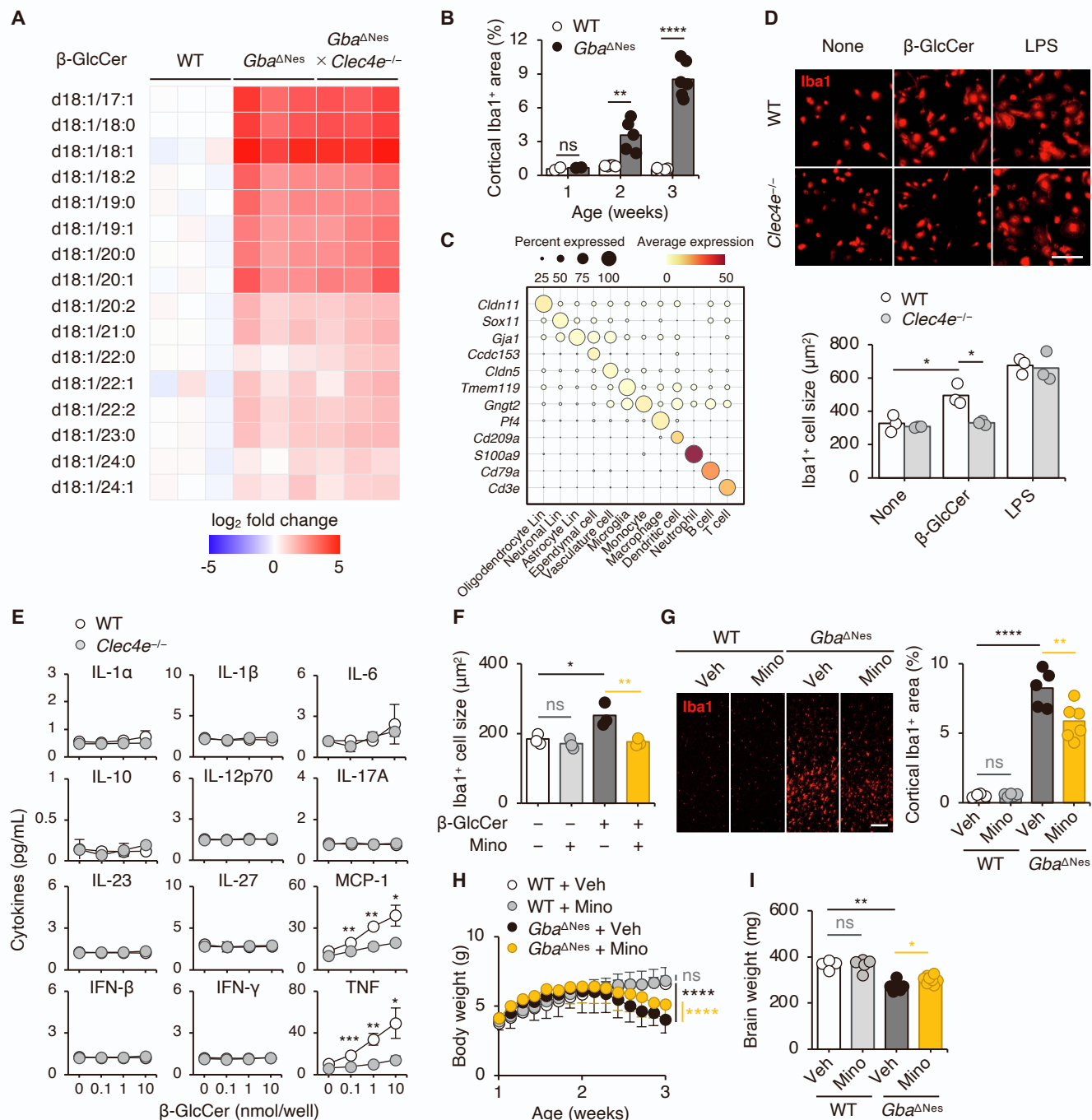


Figure S2. Activation status of microglia in mouse GD models (Related to Figure 2)

(A) Heatmap showing $\log_2$ fold change in the amount of β -GlcCer species in the brain of P14 $Gba^{\Delta Nes}$ and $Gba^{\Delta Nes} \times Clec4e^{-/-}$ mice compared to P14 WT mice ($n = 3$ mice per group). (B) Iba1⁺ area of the cerebral cortex of WT and $Gba^{\Delta Nes}$ mice at indicated age (WT, $n = 2, 5$, and 5 ; $Gba^{\Delta Nes}$, $n = 2, 5$, and 6 for 1, 2, and 3 weeks). (C) Dot plot showing the marker genes for cell types with their percent expressed and average expression. (D) Primary microglia from WT and $Clec4e^{-/-}$ mice were cultured with β -GlcCer d18:1/18:0 (10 nmol/well) and LPS (1 μ g/mL) overnight and stained with Iba1 (red). Iba1⁺ cell size from three technical replicates. (E) Primary microglia from WT and $Clec4e^{-/-}$ mice were cultured overnight with the indicated amount of β -GlcCer d18:1/18:0. Concentration of cytokines in the supernatant from three technical replicates. (F) Primary microglia from WT mice were cultured overnight with or without β -GlcCer d18:1/18:0 (10 nmol/well) and/or Mino (50 μ g/mL). Iba1⁺ cell size from three technical replicates. (G) Iba1 (red) staining of cerebral cortex from P21 $Gba^{\Delta Nes}$ mice with (WT, $n = 5$; $Gba^{\Delta Nes}$, $n = 6$) or without (WT, $n = 4$; $Gba^{\Delta Nes}$, $n = 5$) Mino treatment (left). Iba1⁺ area was quantified (right). (H) Body weight chart for WT and $Gba^{\Delta Nes}$ mice with or without Mino treatment ($n = 5$ mice per group). (I) Brain weight of P21 WT and $Gba^{\Delta Nes}$ mice with (WT, $n = 5$; $Gba^{\Delta Nes}$, $n = 8$) or without (WT, $n = 4$; $Gba^{\Delta Nes}$, $n = 6$) Mino treatment. Scale bars show 30 μ m (D), and 100 μ m (G). Each dot represents an individual mouse (B, G, and I), bars represent the mean of each group (B, D, F, G, and I), and error bars indicate mean $\pm$ standard deviation (E and H). Data are representative of three independent experiments (D and F) and at least three mice per group (G). The one-way ANOVA followed by Tukey–Kramer test (D, F, G, and I), the unpaired two-tailed Student t -test (B and E), and the two-way repeated measures ANOVA followed by Fisher's LSD test (H) were used for statistical analysis. * $p < 0.05$; ** $p < 0.01$; **** $p < 0.0001$; ns, not significant.

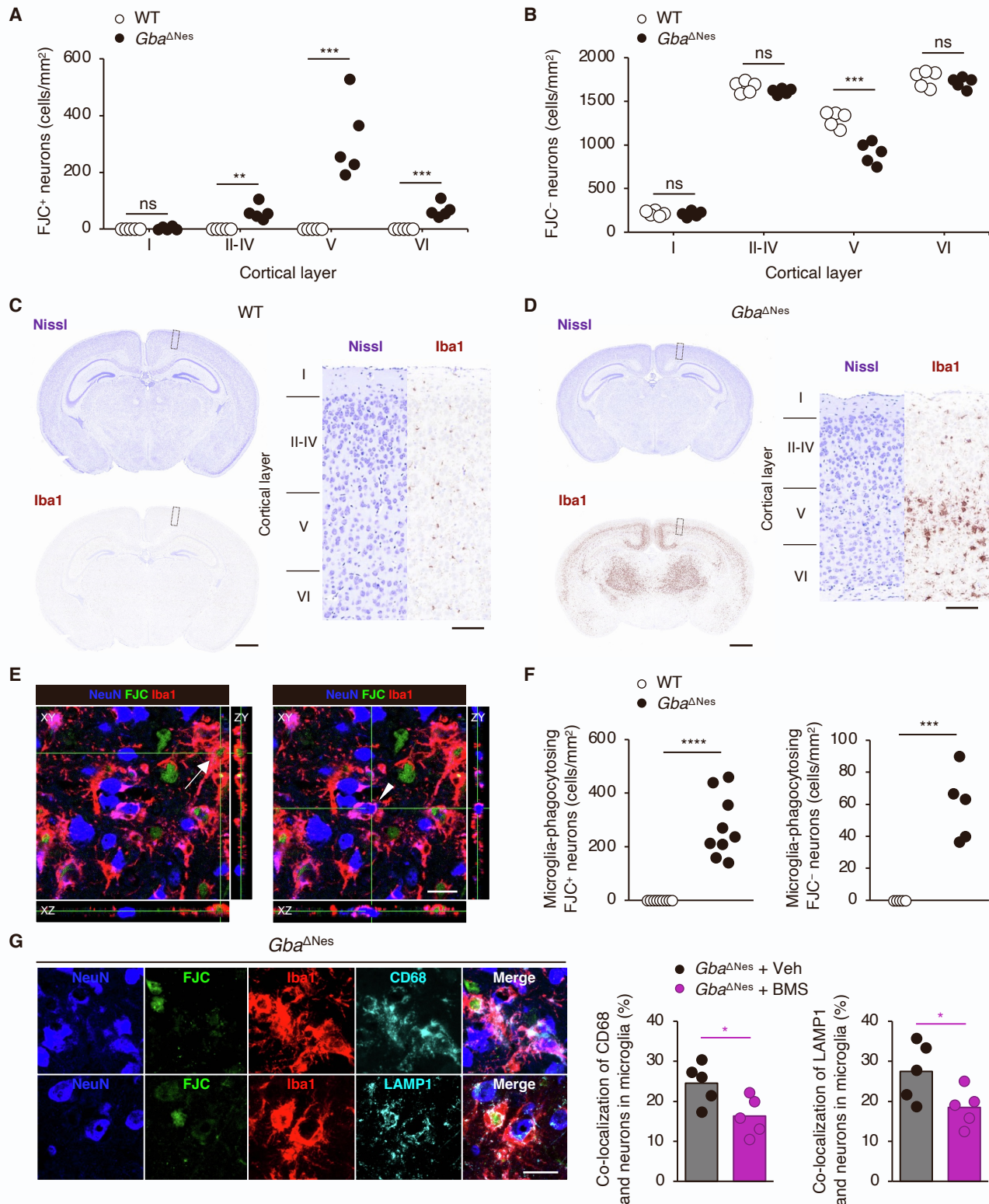


Figure S3. Histological analysis of neuron in *Gba*^{ΔNes} mice (Related to Figure 3)

(A) Number of FJC⁺ neurons of P21 WT and *Gba*^{ΔNes} mice in indicated cortical layers (n = 5 mice per group). (B) Number of FJC⁻ neurons of P21 WT and *Gba*^{ΔNes} mice in indicated cortical layers (n = 5 mice per group). (C) Coronal section from P21 WT mice stained by Iba1 with Nissl counterstain (left). High magnification images showing cortical layers (right). (D) Coronal section from P21 *Gba*^{ΔNes} mice stained by Iba1 with Nissl counterstain (left). High magnification images showing cortical layers (right). (E) Cortical layer V from P21 *Gba*^{ΔNes} mice stained with NeuN (blue), FJC (green), and Iba1 (red) in XY, XZ, and ZY projections. Arrowhead and arrow indicate living neurons (FJC-NeuN⁺) and degenerating neurons (FJC⁺) phagocytosed by microglia, respectively. Analysis was performed on a minimum of 5 mice per group, representative image shown. (F) Number of microglia-phagocytosing FJC⁺ degenerating neurons (n = 9 mice per group) (left) and FJC⁻ living neurons (n = 5 mice per group) (right). (G) Cortical layer V from P21 *Gba*^{ΔNes} mice stained with NeuN (blue), FJC (green), Iba1 (red), and CD68 or LAMP1 (cyan) (left). Percentage of co-localization of CD68 or LAMP1 and neurons in microglia of Veh or BMS-treated *Gba*^{ΔNes} mice (n = 5 mice per group) (right). Scale bars show 1 mm (C and D left panel), 100 μ m (C and D right panel), and 20 μ m (E and G). Each dot represents an independent mouse (A, B, F, and G) and bars are the mean of each group (G). Data were analyzed by the unpaired two-tailed Student *t*-test (A, B, F, and G). **p* < 0.05; ***p* < 0.01; ****p* < 0.001; *****p* < 0.0001; ns, not significant.

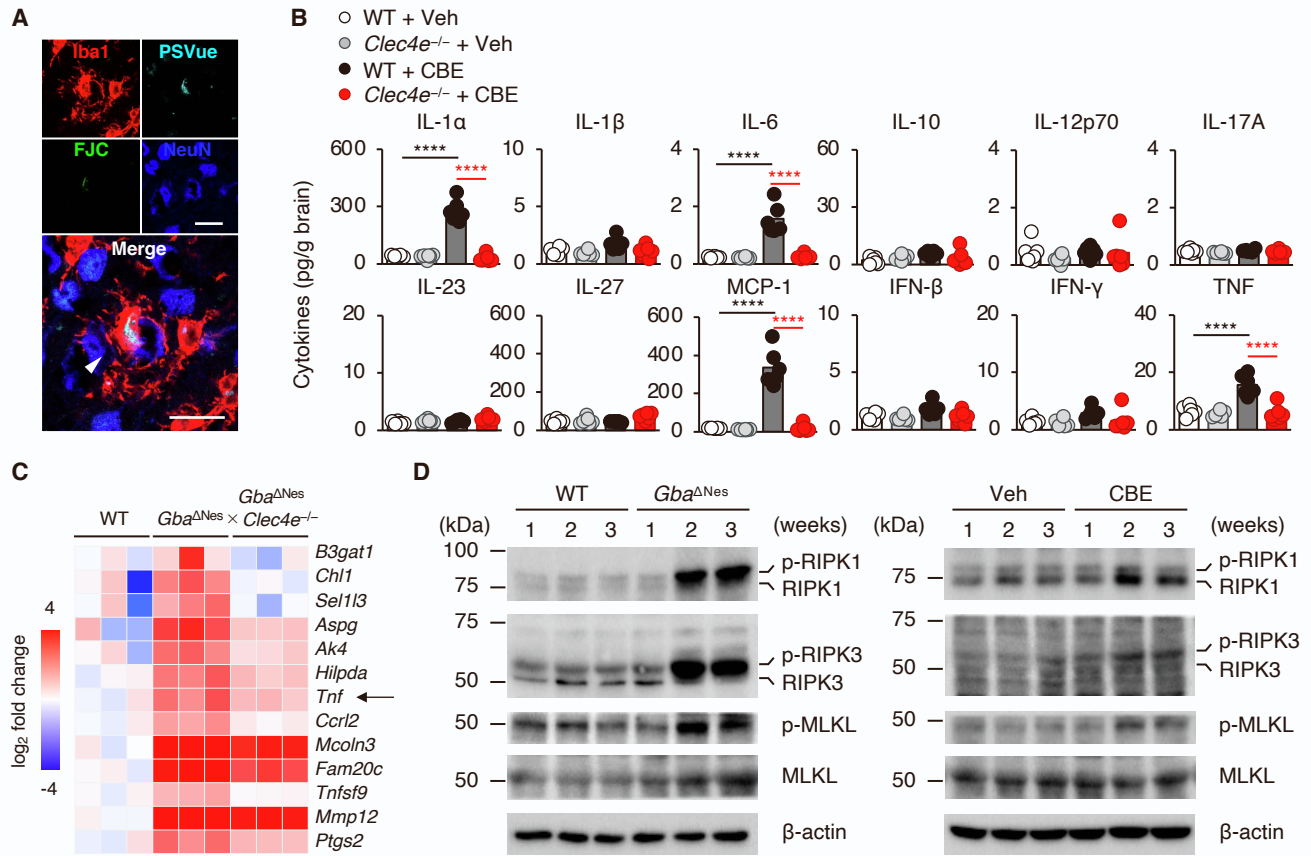


Figure S4. Inflammatory signature in the brain of GD mice (Related to Figure 4)

(A) Cortical layer V from P21 *Gba*^{ΔNes} mice stained with Iba1 (red), PSVue (cyan), FJC (green), and NeuN (blue). Arrowhead indicates PtdSer-exposing living neuron (PSVue+FJC-NeuN⁺) phagocytosed by microglia. Scale bars show 20 μ m. (B) Inflammatory cytokines in the brain from P21 WT and *Clec4e*^{-/-} mice with or without CBE treatment (n = 6 mice per group). Each dot represents an independent mouse and bars are the mean for each group. (C) Bulk RNA-seq of isolated microglia from P14 WT and *Gba*^{ΔNes} mice brain. Heatmap showing log₂ fold change of all downregulated DEGs (p < 0.05, log₂ fold change $\geq$ 2) between *Gba*^{ΔNes} vs. *Gba*^{ΔNes} $\times$ *Clec4e*^{-/-} microglia among upregulated DEGs (p < 0.05, log₂ fold change $\geq$ 2) between WT vs. *Gba*^{ΔNes} microglia (n = 3 mice per group). Arrow indicates *Tnf*. (D) Western blot analysis of phosphorylated and non-phosphorylated RIPK1, RIPK3, and MLKL expression in neurons from a WT and *Gba*^{ΔNes} or Vehicle and CBE-treated WT mice at the indicated age. Data are representative of two mice (A) and two independent experiments (D). The one-way ANOVA followed by Tukey–Kramer test (B) was used for statistical analysis. ****p < 0.0001.

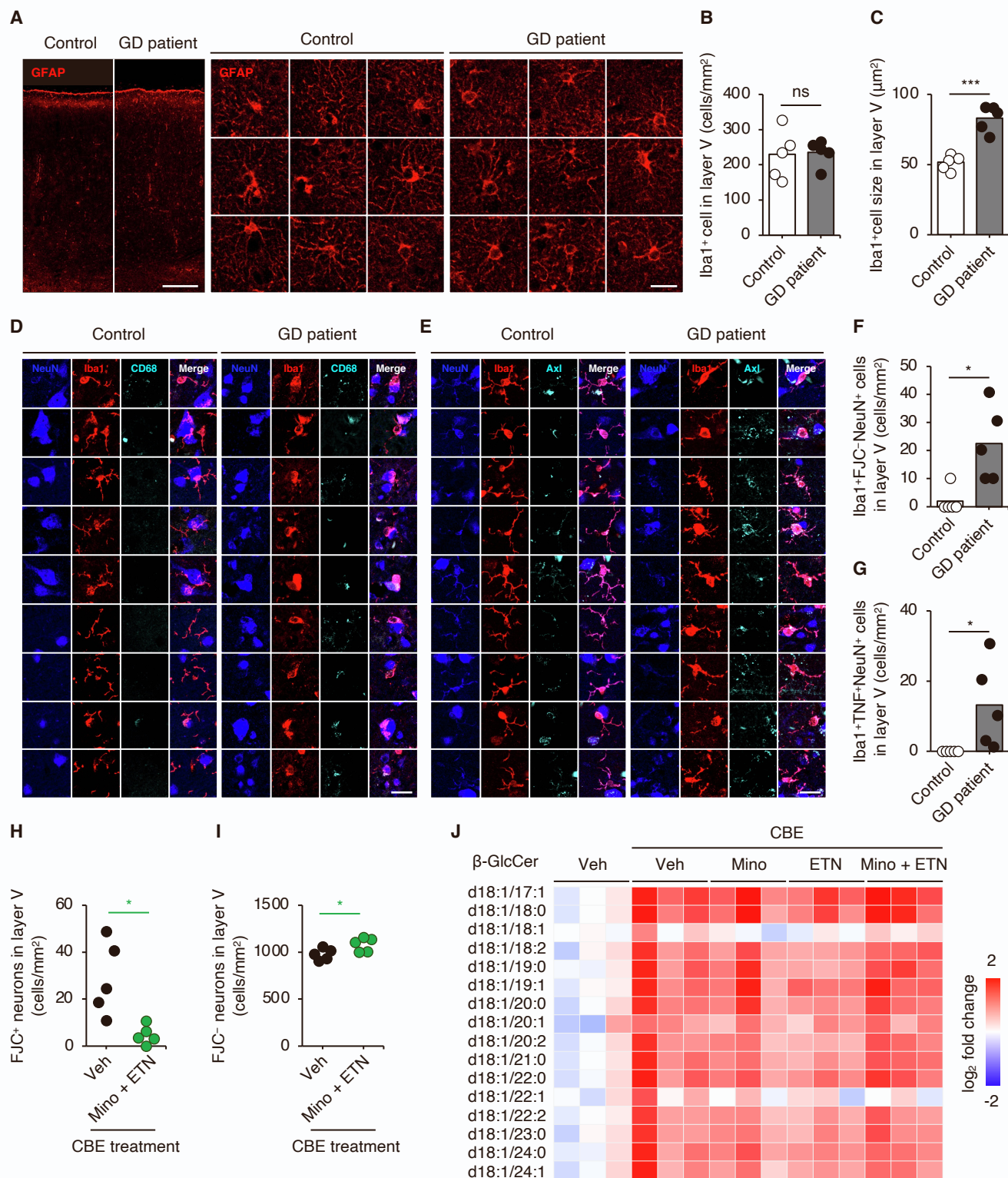


Figure S5. Effect of combinatory use of FDA-approved drugs on neuronal loss (Related to Figure 5)

(A) Images of GFAP⁺ cells in cortical layer V of control and GD patients. (B) Number of Iba1⁺ cells in cortical layer V of control and GD patients. (C) Size of Iba1⁺ cells in cortical layer V of control and GD patients. (D) Images of NeuN (blue), Iba1 (red), and CD68 (cyan) cells in the cortex of control and GD patients. (E) Images of NeuN (blue), Iba1 (red), and Axl (cyan) cells in the cortex of control and GD patients. (F) Number of Iba1⁺ FJC- NeuN⁺ cells in cortical layer V of control and GD patients. (G) Number of Iba1⁺ TNF- NeuN⁺ cells in cortical layer V of control and GD patients. (H) Number of FJC⁺ neurons in layer V of P21 WT mice treated with CBE or CBE + Mino + ETN (n = 5 mice per group). (I) Number of FJC⁺ neurons in layer V of P21 WT mice treated with CBE or CBE + Mino + ETN (n = 5 mice per group). (J) Heatmap showing log₂ fold change in the amount of β-GlcCer species in the brain of P21 CBE mice treated with Veh, Mino, ETN, or Mino + ETN compared with P21 Vehicle-treated WT mice (n = 3 mice per group). Each dot represents an individual patient image (B, C, F, and G) or mouse (H and I). Bars are the mean for each group (B, C, F and G). Scale bars are 500 μm (A left panels) and 20 μm (A right panels, D, and E). Data were analyzed by the unpaired two-tailed Student *t*-test (B, C, and F–I). **p* < 0.05; ****p* < 0.001; ns, not significant.